Natural Knowledge

(high school edition)

spirals, memes, and art

Third edition

Book 2 of *Natural Knowledge - a collection*

Freddi Van Gogh

Natural Knowledge (high school edition): spirals, memes, and art

Third edition

Book 2 of *Natural Knowledge - a collection*

Copyright © 2026 Freddi Van Gogh

First published 2026.

ISBN 979-8-14-075723-6 (paperback)

Licensed under the Creative Commons Attribution-ShareAlike 4.0 International License (CC BY-SA 4.0, https://creativecommons.org/licenses/by-sa/4.0/). Share it, remix it, build on it, sell it — credit the source and release derivative work under the same terms.

Attribution:

> Freddi Van Gogh, *Natural Knowledge (high school edition)* (2026). Licensed CC BY-SA 4.0.

How this book was made

The ideas, the questions and the decisions in this book are the author's. The author wrote the notes, the riffs and the rough drafts, and set out a recipe for each part. AI assistants turned that material into prose, checked it against sources and against the series' fact checker, and raised questions where the notes were wrong or unclear. The author answered them, made the calls, and edited the result.

In the terms publishers now use, the text is AI-generated: an AI drafted it, working from the author's own notes, research and writing as its source material. The cover was also made with an AI assistant, drawn in code.

A visual companion

Want to see these concepts in games and fun video clips? A collection of these and some more detail on visual subjects are further explored at:

https://freddi-van-gogh.github.io/Natural_Knowledge_git/data-vis-compendium/

The code can be found more quickly in the source repository, which also holds the books as PDFs:

https://github.com/freddi-van-gogh/Natural_Knowledge_git

Contents

Start here 1
1. What this is, and what it isn't 5
2. Words to know 11
3. A lot of tries 17
4. Chemistry that keeps itself going 23
5. Bags 29
6. Things inside things 35
7. You are crowded 41
8. Copying 47
9. Evolution leaves tracks 53
10. Spirals 59
11. Time that stacks up 67
12. What counts as alive 71
13. What a brain throws away 77
14. Words 83
15. Knowing without knowing why 93
16. What we don't know 101
17. Where this comes from 111
About the author 135
Natural Knowledge - a collection 137

Start here

This book is about how complicated things come from simple things, without anyone planning it.

It starts with chemistry on a planet with no life on it, and ends with people arguing about what any of it means. In between it covers how a boundary turns a puddle of chemicals into something that can compete, how copying got started, how brains throw away almost everything they receive, and how a cooking technique kept people alive for thousands of years while nobody involved knew what it was doing.

There is a longer version of this book. This one is shorter, uses fewer technical words, and explains things with objects you can picture. It is not a watered-down version in the way that usually means worse. The things that were cut were cut because they were long, not because they were hard.

Where this came from

I should tell you where this started, because it affects how you should read it.

I have a background in biochemistry, computer science, and computational biology. But this book did not

start in a lab. It started during a difficult period when I had a bad reaction to an allergy medication, and I went through weeks of seeing connections everywhere — between science, religion, mythology, language, evolution, and stories. I wrote constantly. I ended up with hundreds of pages of notes.

The experience felt meaningful. It felt, sometimes, like something was guiding it. I am not going to pretend otherwise, and I am also not going to claim that feeling was evidence of anything.

When the episode passed and I went back to the notes, most of what I had written was not usable. But underneath the mess, the same questions kept coming up. Why do some things last and others fall apart? Why do completely different systems sometimes work in similar ways? This book is what was left after I threw out everything I could not check.

That matters because the ability to see patterns is powerful and it is not reliable. You can find a real connection, and you can find one that is not there, and **from the inside they feel exactly the same.** A pattern that feels unmistakably right is the beginning of a question, not the answer to one.

So this book marks which of its claims you should trust, and how much.

How claims are marked

Every book like this mixes three different kinds of statement. Most of them hide that. This one doesn't.

> **Settled.** Scientists agree. You can look it up and find the same answer.

> **Still argued about.** There is real evidence on more than one side, and smart people disagree. The book tells you what the disagreement is instead of picking for you.

> **My own idea.** Something I think, which you should not accept merely because it is printed in a book.

Those appear in boxes like the ones above, and each box covers what's inside it. If a paragraph isn't in a box, it's the first kind — settled enough that saying so every time would get tiring.

If that seems like a strange thing to put in a book, here is why it's there. Almost everything you read presents a conclusion with the arguments cleaned out of it. That makes for smoother reading, and it means you cannot tell the difference between something that is settled and something that is contested but was summarized confidently. Those are extremely different, and you deserve to know which one you are getting.

One more thing before you start

This book does not tell you what it all means.

It explains how chemistry becomes life and how life becomes things that think and talk. It does not tell you whether there is a purpose behind that, whether anything is watching, or what to make of the fact that a bit of matter on one planet ended up able to ask. It is not equipped for those questions, and it says so where they come up.

You can finish this book as a religious person or as an atheist, and nothing in it is designed to move you. That is deliberate. What it is designed to do is show you how each step works, so that whatever you conclude, you conclude it knowing the mechanism.

1
What this is, and what it isn't

Here is the whole argument in one paragraph.

Things vary. Some versions last longer than others. The ones that last get copied, or stick around, or get passed on — and then they vary again. Run that loop long enough, in enough places, and you get complicated things that nobody designed. That is how you get from chemistry on a bare planet to a person reading a sentence about it.

That's it. The rest of the book is showing you each step.

The sequence

A pattern comes up over and over:

Something varies → some versions last → what lasts gets passed on → it varies again

You will see this in chemical reactions, in cells, in animals, in brains, in languages, and in what people cook for dinner. It shows up at every scale.

Start with what's in front of you

A lot of this book is hiding in ordinary places. Mayonnaise is a lesson in how the first cells may have got their walls (chapter 5). A candle flame is a shape that only exists while something keeps flowing through it (chapter 4). A giraffe's neck carries a nerve that takes a ridiculous detour, and the detour is a record of history (chapter 9). And the garden hose coiled up outside is solving the same storage problem as the DNA in every one of your cells (chapter 10).

That's the habit worth building. Look at an everyday shape and ask: *why is it that shape?* Usually there's a real answer, and it's often the same answer as something much bigger.

Usually. Which brings us to the trap.

The trap

Now the warning, and it is the most important thing in this chapter.

The same words fit all those cases. That does not mean the same *thing* is happening.

Take "selection." When a molecule lasts longer than another molecule, nothing is being chosen, nothing is copied, and neither molecule takes anything from the other — one arrangement just falls apart slower. When a rabbit survives and another doesn't, that involves copying, offspring, and generations. When a joke spreads

and another dies, it involves people remembering, getting it slightly wrong, and telling someone else.

Three completely different machines. One word.

The word is still useful, because it points at the same *question*: is something varying, and is something treating the variations unequally? That question is worth asking in all three cases. **The answer is different every time, and the answer is the part that matters.**

If you remember one thing from this book, make it that.

What this is not

It is not a theory. A theory makes predictions that would prove it wrong. This doesn't, yet. What it offers is a set of words that make certain changes easier to describe. That's less than a theory. It isn't nothing.

It does not explain everything with one mechanism. See above.

It does not fill in the gaps. There is a big one, and chapter 8 is where you hit it: nobody knows how chemistry first started copying itself reliably. The book tells you where the hole is instead of walking you past it.

It does not tell you what any of it means. Already said, and worth saying twice.

Why the arguments are left in

One more thing about how this book is built.

When scientists disagree, most books pick the popular answer and present it as settled. This one doesn't. Where there's a real argument, you get the argument.

In chapter 8 there are three competing explanations for something, and I don't pick. In chapter 14 I use a famous experiment and then tell you that when other researchers tried to repeat it, they couldn't. In the last chapter I give you the strongest attack on this entire book, written as well as I can make it, before I respond to it.

That's borrowed from an old tradition of religious legal texts — the Talmud, mainly — where the rulings are written down *along with* the arguments that lost, with names attached, and some questions are marked as unresolved and just left that way. The reasoning behind keeping the losing argument is that someone later might find it was right, and that you should be able to see what a decision was chosen *over*.

The two things are not the same, and I want to be clear about that. Those texts settle questions by authority and tradition. This book has to answer to evidence, and when the evidence comes in, a losing idea doesn't get to stay out of politeness. A failed experiment is not a minority opinion waiting to be vindicated.

What I took is narrower: write down who thought what, keep the losing argument readable, and when a question is genuinely open, leave it open on the page.

Courts and parliaments keep the same kind of record. In many courts, a judge who disagrees with a decision can write a **dissent**, explaining why, and it's published right next to the ruling. Sometimes, years later, another court ends up agreeing with the dissent. Many parliaments keep a word-for-word record of their debates, so the arguments against a law are still there after it passes. They settle things by authority and by vote, though, not by experiment, so the same limits apply to them too.

How this book checks itself

Writing down who thought what is half of it. The other half is making sure the book doesn't contradict itself — and a book this long can contradict itself without anyone noticing. So it comes with a checker, written in a computer language called **Prolog**.

Most computer programs are recipes: do this, then this, then this. Prolog is more like a detective with a notebook. You don't tell it what to do. You give it **facts** — "chapter 9 says a giraffe's nerve is about four and a half metres long," "the longer version of the book says that too" — and **rules** — "if two parts of the book give different numbers for the same thing, that's a problem." Then you ask it a question: *is there a problem anywhere?* It checks every fact against every rule, without getting bored and without skipping the dull ones, and tells you everything it finds.

That's the whole trick. Tell it that all dogs are mammals and that Rex is a dog, and it works out that Rex is a mammal without anyone writing that down. The checker does the same thing with this book. If a conclusion is in a **Settled** box but leans on something that's **Still argued about**, it works out that the conclusion is claiming more than it has earned.

The first time it ran, it found the gap in chapter 8 — nobody knows how chemistry first started copying itself — without being told the gap was there. It also caught a place where the book's own summary stated something more firmly than the chapter it was built on, and that's been fixed. Finding things like that is the point.

It can't tell whether the book is *true*. It can only tell whether the book agrees with itself and with the sources it names. A mistake that's consistent everywhere would get past it, which is a good thing to remember about any checker.

You can read it yourself in *Prolog Logicbase*, book 4 of this collection: the rules it checks, the facts it was given, and how to run it.

2
Words to know

Eight words do most of the work in this book. They're all ordinary words used in a specific way.

Variation — things being different from each other. Two molecules that aren't identical. Two puppies from the same litter. Two retellings of the same story.

Selection — some versions lasting and others not. Careful: this does *not* mean anything is choosing. A sandcastle washing away while a rock stays put is selection in this sense. Nobody picked the rock.

Keeping — once something lasts, something has to hold onto it. A stable shape holds onto it. A copied gene holds onto it. A memory holds onto it. So does a habit nobody questions.

A boundary — anything that separates an inside from an outside. A soap bubble. A cell membrane. Skin. A national border. Chapter 5 is about why boundaries change so much.

Throwing things away — keeping the useful part of something and discarding the rest. Your eye does it. A word does it. A recipe does it. The technical term is *com-*

pression, and the thing to notice is that it always loses something. That's not a flaw. That's the deal.

Passing it on — moving something from one place to another. Parent to offspring. Teacher to student. One cell to the next.

Feedback — when what a system does changes what it does next. A microphone squealing near a speaker is feedback. So is sweating when you get hot.

Emergence — when a group of things together does something none of them does alone. A single bird cannot make a flock turn. A single neuron cannot remember your name.

These aren't explanations

Here's the catch. All eight are *descriptions*.

If I tell you a system "varied and then some versions were selected," I've told you the shape of what happened and deliberately left out the machine that did it. That's useful for noticing that two situations rhyme. It is useless for showing they work the same way.

Watch what happens to one word when you look underneath it:

	What varies	What makes some last	How it's kept
Chemistry	Molecules bumping into each other	Which arrangements survive the next knock	Nothing is copied. It lasts or it doesn't.
Living things	Genes, through mutation	Which individuals survive and reproduce	A gene is copied into offspring, very accurately
Ideas	Retellings, misremembering, invention	Whether people notice it, get it, and repeat it	Rebuilt from scratch by someone who may have misunderstood

Three rows, one heading. **No shared machinery whatsoever.**

Chemistry has no population and no copying. Living things need a gene and generations. Ideas spread between strangers, in any direction, including from a kid to a grandparent, and they get copied so sloppily that what arrives is often not what was sent.

So when you see "selection" in this book, read it as a question, not an answer. The question is: *is something varying, and is something treating the variations unequally?* Then go find out how, because the how is different every time.

A shorthand for mixing ideas

This book has a companion called *Modern Alchemy*, book 3 of this collection. It gives each idea a short sym-

bol, the way chemistry gives each element a symbol: Ap for molecules with a water-loving end and a water-hating end (chapter 5 is about them), Df for things drifting from crowded places to emptier ones, V for a closed bag. Then it adds a few signs for combining them. + means "these act together." → means "can produce, if the conditions are right." So

Ap + Df → V

says that those molecules, drifting around in water, can close up into a bag. A chapter's worth of chemistry, in a single line. Put lines from different chapters side by side and you can look for ones that rhyme, swap a piece, and ask what would happen. That's the fun of it, and it's a real way to get ideas.

Why call it alchemy? Because alchemists did exactly this. They had symbols for substances and for processes, and they wrote recipes in them, sure that the right combination would reveal how matter works. They were wrong about almost everything, and chemistry grew out of their workshops anyway.

Here's the catch, and it's the same catch the alchemists hit. Symbols always combine. A recipe for turning lead into gold looks as neat on paper as a recipe for bread, and the paper won't complain. So a formula in *Modern Alchemy* is a question, not an answer: it means "this might be worth checking," never "this is how it works." There's a separate sign, ≈, for "looks alike, but don't assume the same thing is happening," and it's kept

away from the arrow on purpose. And no formula can show how long something takes, how likely it is, or all the times it failed. It only writes down what lasted.

3
A lot of tries

"It took billions of years."

You've heard that about the origin of life. It's true and it explains almost nothing, because nobody can picture a billion years. The number is so big it stops being information and turns into a way of ending the conversation.

Here's a better question. Not *how long did it take*. **How many times did it happen?**

One puddle

Picture a shallow pool at the edge of an ocean on the early Earth. Waves wash over it. It partly dries in the sun. It fills again.

In that water are molecules with a split personality: one end likes water, the other end hates it. (Soap is like this. So is the fat in your cell membranes.) When the pool gets stirred, those molecules clump into films and little bubbles. The bubbles hold together for a while. Then the next wave smashes them.

Each one of those bubbles is a try. Slightly different mix of chemicals inside, held together for a slightly different length of time, under slightly different conditions.

Now suppose the pool gets stirred and resettles every six minutes. Count:

	tries
in a day	about 240
in a year	about 88,000
in a century	about 8.8 million
in ten million years	about a trillion

Ten million years is nothing in Earth's history. Geologists would barely give it a name. **And that's one puddle.**

Does the number matter?

I made up the six minutes, and I should say so, because you should always be suspicious of a number someone picked.

So let's break it. Say the pool only resets once an hour. That cuts the total by ten — about ninety billion. Say once a *day*. That cuts it by 240 — about four billion.

Four billion tries. In one puddle. In a blink of geological time, using the most pessimistic number I can defend.

That's the point worth noticing: **the argument doesn't depend on my number.** Pick any reasonable cycle and you get the same conclusion. When an argu-

ment survives you attacking its weakest part, that's a sign it's resting on the situation rather than on the arithmetic.

And there wasn't one puddle. There were tidal flats, hot springs, volcanic vents, wet cracks in rock, films on mineral surfaces — the whole planet, all at once, for hundreds of millions of years.

> **My own idea.** So the picture isn't one slow process grinding away forever. It's an absurd number of short, cheap, parallel experiments, almost all of which failed within minutes, running everywhere at the same time. Framing it this way — counting tries instead of years — is mine, and so are the numbers in that table, which are an illustration and not a measurement.

The pieces underneath it are solid: wet–dry cycles really do drive chemistry, those split-personality molecules really do assemble on their own, and both have been demonstrated in labs many times.

Not random, though

There's a famous image: enough monkeys at enough typewriters will eventually type Shakespeare.

It's a good way to feel how big numbers make unlikely things ordinary. It's also wrong in one way that matters.

Chemistry is not a monkey. A monkey can hit any key. A molecule can't do just anything — bond strengths, temperature, what's nearby, and what shape things are all limit what can happen next. Most possible arrangements of a set of atoms are never tried at all, because there's no path to them from where the system is.

That's good news, not bad. A process that tried everything at random would need far more time than Earth has had. A process whose next move is heavily restricted by what's already there is searching a much smaller space — and searching it in parallel, everywhere, constantly.

The thing you can't see from here

One warning before we move on, and it applies to this whole book.

You can look at the chemistry that led to life. You cannot look at the chemistry that didn't, because it left nothing behind.

That's a problem called **selection bias** — you're only seeing the survivors. It's especially bad here, because the one path that worked is the only one that eventually produced anybody capable of wondering about it.

So: the fact that it happened doesn't tell you it was likely. It doesn't tell you it was inevitable, either. A planet running an absurd number of experiments will

produce *something* improbable, and we are standing inside the one it produced.

How likely was life, given early Earth? Genuinely nobody knows. We have one example.

4
Chemistry that keeps itself going

Most chemical reactions happen once and stop. Some don't. This chapter is about the ones that don't, because that's the first step towards anything that could be called alive.

A reaction that builds its own helper

A **catalyst** is something that makes a reaction go faster without being used up. Your body is full of them.

Now imagine a reaction whose *product* is a catalyst for that same reaction. It makes the thing that makes it go faster. So it starts slow, speeds up as the product builds, and then takes off.

This is real, it's been known since 1861, and it uses a chemical the early Earth almost certainly made, though maybe not in the amounts the reaction needs. Formaldehyde — a simple molecule — reacting with itself in water can produce sugars. One of the first things it makes helps make more of it. The reaction crawls along for ages doing nothing much, and then accelerates hard.

That shape — slow, slow, slow, *fast* — is the fingerprint. It's the same shape as a rumour spreading or a population growing.

How long is long enough?

Why does speed matter so much? Because nothing holds still forever. A reaction that takes longer than its puddle lasts might as well not happen.

So how long *does* a place hold still? That depends on what you call a place. An environment isn't something nature hands you; it's an idea people use, and you can draw it at any size. Your face is an environment for the bacteria living on it. The shady corner of your yard is a microenvironment, cooler and damper than the sunny patch a few steps away. Your house is an environment, and so are your town, your region, and all of nature. The edges go wherever you decide to draw them.

Time is as loose as the edges. There's no single right timescale, only the question of whether there was enough time for something to happen. A tide pool might hold still for minutes or hours, between waves and tides. Water sealed in a crack deep underground can sit for far longer. A catalyst matters because it lets a reaction finish inside whatever window it gets.

A molecule that makes copies of itself

Better: in 1996 researchers built a small protein, 32 building blocks long, that makes copies of itself. Give it

the two halves of itself and it grabs them, holds them against its own body, and helps them join into a complete new copy.

No cell. No life. A molecule, in a tube, making more of itself.

Which is amazing, and it comes with a catch that matters later.

The copying gets *slower per copy* as copies pile up, because copies stick to each other, and a copy stuck to another copy can't do any work. That sounds like a technicality. It isn't. It means a faster version can never wipe out a slower one — the faster one's advantage shrinks exactly as it becomes common.

And "faster versions outcompeting slower ones" is the engine of evolution. Without it, nothing improves over generations. This molecule copies itself and goes nowhere.

Almost every self-copying chemical system built since has the same problem.

The newest one, reported in 2026, is different in one way: it didn't start as a design. Researchers fished it out of a huge pool of random RNA sequences, then improved it in the lab. It's only 45 letters long, and it can build both strands a copy needs — itself and its matching partner. But it builds them in separate reactions, it gets about one letter in seventeen wrong, and after 72 days only about two copies in a thousand are finished. Its discoverers think molecules like it may be more com-

mon than anyone guessed. Nobody has yet got such a molecule copying itself on its own, generation after generation.

A loop with no single hero

There's another route, and it may be more realistic, because it doesn't need any single molecule to be clever.

Suppose A helps make B, B helps make C, and C helps make A. No molecule makes itself. **The set makes itself.** Feed it raw material and it keeps going, and the ability to keep going belongs to the loop, not to any member of it.

The maths here is better than the chemistry. It's been shown that in a big enough, messy enough mixture, loops like this should turn up almost unavoidably. The lab versions that exist are small and mostly built on purpose rather than found.

> **Still argued about.** Whether real early-Earth chemistry was big and messy *enough*, and whether the catalysts available were good enough. Nobody knows yet.

Shapes that only exist while something flows

Last idea, and it's the one that explains why any of this sticks around.

Heat a shallow pan of oil evenly from below. For a while, nothing. Then past a certain temperature the

whole thing suddenly organizes itself into a tidy grid of circulating cells, often six-sided, like a honeycomb. Nobody arranged them.

Turn off the heat. The pattern doesn't freeze. **It vanishes.**

That's the thing to hold onto. Some shapes aren't objects. They're processes that look like objects as long as energy keeps moving through them. A flame is one. A whirlpool is one. A hurricane is one.

So is every living thing. You are not an object that was assembled and now sits there. You are a set of processes rebuilding you continuously, and you last exactly as long as they keep running.

What this does not solve

Three honest problems, because a chapter with only good news isn't worth much.

The sugar reaction is a mess. It doesn't make one thing. It makes a sludge of sugars and junk — the literature calls it tar, fairly — and the sugar you'd actually want for life is a small fraction of it and falls apart easily. The problem isn't getting the reaction to go. It's getting it to be *picky*.

The self-copying molecules were designed, or picked out and improved, by chemists. Showing a chemist can build or find one is not showing that chemistry can stumble onto one.

Nothing here evolves. This is the big one, and it's the parabolic problem from earlier, generalized. These systems keep going and they copy themselves. What nobody has built is a chemical system where better versions keep taking over from worse ones, so improvements pile up without limit.

That's the line between persistent chemistry and life. We'll hit it again in chapter 8, and it's still the biggest hole in this whole account.

5
Bags

Shake oil and vinegar in a jar. They mix, they look cloudy, and within a couple of minutes they've separated again. Oil and water don't mix, and shaking only delays the inevitable.

Now add an egg yolk and whisk. You get mayonnaise, and mayonnaise does not separate.

What changed?

Molecules with two personalities

The yolk brought in molecules with a split personality: one end is attracted to water, the other end is repelled by it.

A molecule like that has nowhere comfortable to sit in a mix of oil and water — except right at the boundary between them, with each end in the stuff it likes. So that's where they go. Enough of them line up around each oil droplet, and the droplets can't merge anymore.

Nobody arranged them. Each molecule found the spot where it was most comfortable, and a stable structure is what the sum of all those spots looks like from outside.

That's the whole trick, and some version of it is behind every cell in your body.

Not every bag has an inside

Soap does the same thing in reverse. In water, soap molecules bunch up with their water-hating ends tucked inward and their water-loving ends facing out. You get a little ball with a greasy middle. Grease dissolves into it, which is why soap cleans.

Here's a distinction that's easy to miss and matters a lot. **That ball has no inside.** Its middle is a tangle of greasy tails with no room for water. You can't put anything watery in there.

The structure that *does* have an inside is a **double layer**: two sheets of these molecules back to back, greasy tails pointed at each other in the middle, water-friendly ends facing out on both faces. Curl that around and close it up and you have a bag — a pocket of water, sealed off from the water outside by a film two molecules thick.

Which one you get depends on the shape of the molecule. One greasy tail, and they pack into solid balls. Two tails, and they can't taper to a point, so they form sheets instead — and sheets close into bags. The molecules in your cell membranes have two tails.

This matters because the simple molecules are the believable ones for early Earth. Fatty acids are short, have one tail, turn up easily in lab conditions meant to

mimic the early planet, and have even been found in meteorites. Under the right conditions they make bags too. In the lab, those bags grow, split when squeezed through something narrow, and trap bigger molecules inside while they do it.

Why a bag changes everything

Without a bag, a useful reaction happens and its products drift away. Nothing benefits.

With a bag, the products stay with the system that made them. And — this is the part that matters — bags can now be **compared**. One bag can last longer than another.

That doesn't need genes, heredity, or life. It needs variation between bags and a difference in how long they last. Which is exactly the minimum condition from chapter 2.

Bags that eat other bags

Here's a real experiment, because this part deserves more than an assertion.

Fatty acid bags constantly trade molecules with the water around them and with each other. Molecules leave one membrane and join another.

Now put more stuff inside one bag. More stuff inside means more pressure pushing outward, which puts the membrane under tension — stretched, like an over-

inflated balloon. **And a stretched membrane grabs new molecules faster than a relaxed one.**

So in a dish of these bags, the full ones grow. And they grow by taking membrane molecules that the empty ones lose. The full bags swell. The empty ones shrink.

Nothing is copied. Nothing is inherited. No organism is involved. A bag that happens to contain more is better at collecting the material it's made of — and it does that at its neighbours' expense.

That's competition, between things that are not alive, driven by nothing but pressure and the physics of a stretched film.

It's about as close as plain chemistry gets to natural selection, and it shows up early, out of nothing more than the shape of a molecule and the behaviour of water.

One more thing a membrane does

A membrane can also hold a difference between its two sides — more of some charged particle inside than outside.

Think of a water wheel. The wheel doesn't make energy. It takes energy out of water falling from a height that something else maintains. A membrane holding more of something on one side is maintaining a difference in the same sort of way, and things in the

membrane can be driven by letting that difference even out.

This is how your cells power themselves, and how plants do photosynthesis. Both work by pumping particles across a membrane and then harvesting the flow back. Picture someone carrying buckets of water back up to the top of the stream so the wheel keeps turning. The pumping is the carrying, and the energy for it comes from food, or from sunlight in a plant.

The comparison is useful. The conclusion that early bags had any of that machinery is not — modern cell energetics is elaborate and evolved. What the earliest bags plausibly had was the simpler thing underneath: a boundary across which a difference could exist at all.

Still argued about. Some researchers flip the order entirely and start from differences that were already there — the natural gaps in acidity across the rock walls of deep-sea vents, where the planet supplies the gradient and the first membranes only have to take advantage of it.

6
Things inside things

Once you have a bag, you can put bags inside bags.

Molecules group into networks. Networks sit inside compartments. Compartments become cells. Cells form tissues. Tissues form organs. Organs form you. You live in an ecosystem.

At every level the bigger thing can do something none of its parts can do alone.

Specializing costs you

When parts of a system each get good at one job, the whole system gets better. Nobody has to be mediocre at everything.

But there's a bill, and it comes due immediately. **A part that gets very good at one job usually loses the ability to survive on its own.**

A liver cell cannot live as a free-swimming organism. It has no idea how. It gave that up, permanently, in exchange for being extremely good at being a liver cell.

So the whole thing now has to coordinate its parts, move resources between them, keep conditions stable,

and manage the conflicts between what's good for one part and what's good for the whole.

That last one turns out to be the interesting problem.

The big jumps

Biologists have a name for the handful of moments when this happened at a large scale: **major transitions**. The list includes loose molecules getting enclosed in compartments, separate genes getting linked into chromosomes, single cells becoming multicellular bodies, and individual insects becoming colonies where most members never reproduce.

Every one has the same signature. Before the transition, the pieces could reproduce on their own. After it, they can only reproduce as part of something bigger.

> **Still argued about.** Which events belong on that list is genuinely disputed. Some researchers think the list was assembled because each item is big rather than because they share a mechanism.

The signature is the part that survives the argument. Something gets given up each time, and it's always the same thing: **the ability to go it alone.**

The best example

The clearest case is sitting in your cells right now.

The mitochondria that release energy in every one of your cells used to be free-living bacteria. They were taken up by a bigger cell, and they never left.

The evidence is unusually direct. Mitochondria have their own DNA, separate from the DNA in your cell nucleus, and it's a loop, the way bacterial DNA is. They have their own protein-building machinery, and it's bacterial-style, close enough that some antibiotics aimed at bacteria also hit them. They're wrapped in a double membrane, which is what you'd expect of something swallowed. They don't get built from scratch — they split in two, like bacteria.

The paper that made the case properly was rejected by around fifteen journals before one published it, in 1967. It stayed on the fringe for years. Then DNA sequencing arrived and it became very hard to argue with. It's now in every textbook.

That's worth remembering for two reasons, and people usually only take the first. Yes: a field being confident doesn't mean a field is right. **Also:** a theory being rejected fifteen times and later vindicated is a real pattern, and it tells you nothing about any *particular* rejected theory. Most ideas that fifteen journals turn down were turned down correctly.

Where the genes went

Here's the detail that makes it sink in.

The original bacterium would have carried a few thousand genes. Your mitochondria have **thirty-seven**.

The rest weren't thrown away. Almost all of them moved into your cell's nucleus, where they still are — still making the proteins the mitochondria need, which are then built in the main cell and shipped back in.

So the mitochondrion didn't just become dependent. It handed over the blueprints for itself. That's not two partners cooperating. That's one system that can no longer be taken apart into the things it was made from.

The part nobody mentions

Everything above describes this as something that happened and then stayed put. It does not stay put by itself.

Your body is a deal between about thirty trillion cells. Every one of them that has a nucleus still carries the machinery to divide — machinery that's a billion years older than animals. The deal only works because that machinery is held down: by signals telling cells when to stop, by cells destroying themselves when they malfunction, by an immune system checking.

Cancer is what it looks like when that fails.

A line of cells goes back to doing what single cells do — dividing whenever conditions allow, with no regard for the body it's in. It isn't an invader. It's a part of you reverting to the level below, using an ability it never lost.

The same tension shows up elsewhere. There are stretches of DNA that are good at getting themselves copied and do nothing for the organism carrying them. In ant and bee colonies, workers actively police each other's attempts to reproduce.

The general shape: **a higher level only survives while it suppresses the level below, and the suppression is never complete.**

Which is the same thing as the flame in chapter 4, one size up. A flame exists only while fuel keeps burning. A body exists only while it keeps holding its own cells down. Neither one is a thing that got built and now sits there. Both are processes that have to keep running.

7
You are crowded

You are not just made of your own cells. You are covered in and filled with other organisms — bacteria, fungi, viruses, and more — living on your skin, in your mouth, and in enormous numbers in your gut.

They are not passengers. Some of them are doing jobs you need done.

The gut

Gut bacteria break down food your own enzymes can't handle. They make compounds your body uses. They help train your immune system. They take up space that would otherwise go to something harmful. They change how some medicines work.

And they do it as a *group*. One species makes something another species eats. One pathway's waste is another's raw material. What the community does depends on how its members interact, not on what any one of them can do.

That sounds like the specialization from the last chapter, and it is — with one big difference. Your cells share your DNA and share your fate. These organisms

don't. They're separate, with their own interests, and the relationship includes cooperation, competition, and outright exploitation depending on conditions.

Skin is not a wall

Skin makes the point more sharply, because skin is where you and the microbial world actually negotiate.

Start with the history. When the first vertebrates crawled onto land, bacteria had already been covering every available surface for *billions* of years. Skin didn't evolve first and then get colonized. **Skin evolved into an already-occupied world.**

That changes what it's for. The outer layer of your skin is often described as bricks and mortar: flattened dead cells packed in a greasy matrix. The obvious reading is a wall to keep things out. A better reading is a surface tuned over a very long time to be livable for certain organisms and not others.

Look at what those cells give up to do it. As a skin cell moves towards the surface, it destroys its own nucleus and internal machinery and ends as a protein-filled husk. It can't divide, repair itself, or react to anything.

It also can't be hijacked by a virus that needs working cell machinery, and it can't accumulate sun damage in DNA it no longer has.

The cell trades every single thing it could do for the one property the barrier needs: being inert.

Your skin publishes a job listing

Your skin doesn't build its microbiome. It sets conditions — how acidic, how oily, how wet, how salty — and whichever organisms can tolerate those conditions move in.

That's the tide pool from chapter 3, on a different surface. Produce an environment, and let whatever survives, survive. **Nothing is choosing.**

And the conditions aren't the same everywhere. Your armpit, your forehead, and your forearm are different worlds — different acidity, different oils, different moisture — with different communities in each. Two spots on your own body can differ from each other more than the same spot differs between you and a stranger.

The second thing is stranger. Your outer skin is completely rebuilt from underneath every few weeks. **The skin on your hand right now was not there a month ago and will be gone a month from now.** It's a flame, in the chapter 4 sense: it exists only because it never stops being remade. Stop the turnover and you don't get a stable barrier — you get one that fails the first time anything tests it.

And it runs both ways. Animals raised in total sterility don't develop normal skin immunity. Particular bacteria have been shown to tune your immune responses, locally, independently of anything happening in your

gut. The microbes aren't tolerated by the barrier. **They help build it.**

A jar of mud

Here's the job-listing idea in a jar. Science classes do this: fill a clear jar with pond mud, mix in some dead leaves and a little calcium sulfate, and leave it in the light. No bacteria are added.

In one study, the bottom turned black within the first week. Down there, where there's no oxygen, bacteria use sulfate the way you use oxygen, and give off hydrogen sulfide, the gas that smells of rotten eggs. It reacts with iron in the mud and makes a black mineral. Near the top, patches of rusty orange can show up within days. Over the following weeks, bands of colour can appear in between: green, and higher up, reddish purple, made by bacteria that use that gas for photosynthesis. A film you can see grows on the inside of the jar, but only in the light.

The bands don't always appear. Nobody put any of these bacteria there; they were already in the mud. In one study, most of the film was made of kinds that had been rare in the mud at the start. **Nothing is choosing.**

How anyone knows this

Quick detour into method, because the method is itself an example of this book's subject.

Almost everything known about these communities comes from reading one gene — the one for a part of the bacterial ribosome, the machine that builds proteins.

That gene has a strange structure. Along its length, stretches that are nearly identical in *every* bacterium alternate with stretches that differ wildly between species.

Neither is an accident. The identical stretches are the parts that do the actual work of building proteins, and building proteins is not optional — a bacterium that mutates there doesn't have descendants. The variable stretches are the parts where the exact sequence matters less, and over billions of years they've drifted.

So one molecule carries two records at once: **what could never be allowed to change, and what there was never any reason to protect.** The gene is a map of its own constraints.

That's also what makes it useful. The identical parts give you something to grab onto that works on any bacterium, including ones nobody has ever seen. The variable parts in between tell you which bacterium it was. You can identify organisms without ever growing them — which matters, because most of them can't be grown.

It has real limits, and a chapter that explains a method and hides its weaknesses isn't worth much. Different species carry different numbers of copies of the gene, which skews abundance counts. You often can't tell closely related species apart. The amplification step

introduces biases of its own. And it only sees bacteria and archaea — the fungi and viruses in the same community need completely different approaches.

8
Copying

Everything so far — chemistry that sustains itself, bags that compete — happens without anything being copied.

This chapter is about copying, which changes everything. It's also where the biggest hole in this book is, so read to the end.

The code

Your DNA stores instructions for building proteins. The translation between them is called the **genetic code**, and it's worth looking at closely.

The mechanics are simple. DNA is written in four letters. They're read three at a time. Four letters in three slots gives 64 possible triplets. Those 64 triplets specify 20 amino acids — the building blocks of proteins — plus a "stop here" instruction.

64 triplets, 21 meanings. So there's slack: several different triplets mean the same thing.

And the slack isn't spread evenly. That's the interesting part.

Most of it is in the third letter. For a lot of amino acids, all four options in the third slot mean the same

thing. A copying mistake there often changes nothing at all about the protein that gets built.

The first two letters carry most of the meaning. A mistake there usually does change the amino acid — but not randomly. Triplets that differ in the first letter tend to code for amino acids with *similar chemical properties*. So an error is more likely to swap one greasy building block for another greasy one, and proteins can often shrug that off.

The start and stop instructions are the least forgiving. Break those and you've damaged the whole protein, not one spot in it.

So: **where a mistake would be cheap, the code absorbs it. Where a mistake would be expensive, it doesn't.**

Throwing away the right things

That's exactly what good compression looks like. Discard what you can lose without wrecking the message. Protect what you can't.

And the code is measurably good at it. In 1998 two researchers compared the real code against enormous numbers of randomly scrambled alternative codes, scoring each by how much damage a typical error would do. Only about **one in a million** random codes did better than the real one. The exact figure depends on how you score the damage, and it's been argued over ever since.

What has held up is the main result: the real code is far better than a typical random code.

Still argued about. Why it's like that is genuinely unsettled, and there are three serious answers.

It was selected for. Lineages whose codes handled errors more gracefully left more descendants, so the code got better at it over time.

It's a frozen accident. The assignments were mostly arbitrary, locked in very early, and became impossible to change — because changing what one triplet means rewrites every protein in the organism at once. Any error tolerance would be a side effect.

It fell out of how the code grew. If the code was built up by adding amino acids one at a time, with each newcomer getting triplets near a chemically similar amino acid already in the code, then error tolerance appears *on its own*. Nobody selected for it. Nothing optimized anything.

Hold onto that third one. It describes something that looks designed, behaves as if it were designed, and was never designed for anything — the appearance falling out of nothing but the order the parts arrived in.

That is the single thing this entire book is most at risk of getting wrong, everywhere, about everything. Keep it in mind for the rest of the chapters.

What *isn't* argued about is the structure itself. However it got that way, the code treats cheap mistakes and expensive mistakes differently.

One comparison, carefully

This resembles error-correcting codes in computing — the thing that lets a scratched disc still play. Worth noticing, and worth not overstating.

An engineered code adds extra check-data on purpose so errors can be spotted and undone. The genetic code does nothing of the kind. It survives errors because of how the mapping happens to be laid out, not because it's checking anything, and it can't detect or fix a thing on its own.

Same kind of behaviour. Completely different means. That's a resemblance, not a shared mechanism — which is the warning from chapter 1, showing up right on schedule.

The hole

Copying with occasional mistakes, in a population where some copies do better, will change across generations with nothing steering it. That's evolution, and once you have it, everything in chapters 9 onward follows.

Here's the problem.

Nobody knows how it started.

Chapters 3 through 5 describe chemistry that organizes itself, concentrates itself, and sustains itself — all demonstrated in labs. Everything after this chapter assumes copying already works. How a chemical system first came to copy a pattern reliably enough for differences to accumulate is not known, and nothing in this book narrows it down. The closest anyone has come is the 45-letter RNA from chapter 4, which can make itself and its partner strand, slowly and with errors, in separate steps. Even its discoverers write that nobody knows how copying first began.

That is not a detail waiting to be filled in. It is the central transition, and this account cannot describe it.

I'd rather tell you that in the middle of the book than bury it at the end.

9
Evolution leaves tracks

Once things copy themselves with variation, and some copies do better, you get evolution. Traits are inherited. Copying introduces mistakes. Some versions survive and reproduce more. Chance shuffles things too, especially in small populations.

None of that aims at anything. Evolution has no destination and no push towards complexity. By any measure that counts individuals, the most successful organisms on Earth are single-celled and always have been. Complicated structures get *lost* all the time when they stop paying for themselves — whales lost their back legs, cave fish lose their eyes, parasites throw away whole organ systems their host can supply instead.

What evolution does leave behind is a record. Not just of what worked, but of **the route taken to get there.** Two kinds of evidence show this, and they point in opposite directions.

Same answer, over and over

Unrelated animals facing the same problem often land on the same solution.

Eyes are the standard case. Light-sensing structures have appeared independently many times across the animal kingdom — surveys put it somewhere above forty, depending on how strictly you count. Vertebrates built a camera eye with one lens. Insects built a compound eye out of many tiny units. Octopuses built a camera eye too, completely separately from us.

Oxygen transport is the same story. Your blood carries oxygen on iron. Many snails and crabs use a totally unrelated protein that carries it on copper instead. Different molecules, arrived at independently, doing the same job.

Powered flight evolved four separate times: insects, pterosaurs, birds, bats. Four different starting bodies.

This isn't coincidence and it isn't a tendency towards anything. **The physics is the same everywhere.** Focusing light, grabbing oxygen and letting go of it again, pushing air down hard enough to stay up — each of those problems allows only a limited range of workable answers. Evolution doesn't invent from nothing. It finds its way into the set of things physics permits, and when the same problem comes up it keeps finding the same region.

Same route, still visible

That's what the constraints allow. Here's what history forces.

Your voice box is wired through your chest.

There's a nerve running from your brain to your larynx, in your throat. It does not go straight there. It goes down into your chest, loops underneath a major artery near your heart, and comes back up to a destination a few centimetres from where it started.

In a giraffe, that same loop goes all the way down the neck and back up. **About four and a half metres of nerve, to cross a gap of inches.**

Here's why. In fish, that nerve runs a short, direct path to the gill structure it serves. As the ancestors of land animals evolved, the heart and its big blood vessels shifted back into the chest — and the nerve, already hooked around one of those vessels, got dragged along. Every individual step was small enough to live with. No step could undo it without disconnecting the nerve. So the detour stayed, and got longer every time a neck did.

Think of an old city. Its streets were laid out for a much smaller town, and as the city grew it built onto them instead of starting over, so getting across town can mean a strange detour, a few blocks longer than walking straight there. Nobody planned the detour. Each new street made sense when it was built. The nerve's route is the same kind of leftover, and the long way round costs little enough that living with it was never a problem.

Your ear used to be your jaw

Two of the three tiny bones that carry sound from your eardrum to your inner ear were, in your distant ancestors, part of the **jaw joint**.

The obvious objection is obvious: how? An animal can't stop being able to chew for a few million years while its jaw turns into an ear.

It didn't have to. Some early mammal relatives had **two working jaw joints at the same time** — the old one, and a new one right next to it made from different bones. With the new joint taking the load, the old one was free to be repurposed. There was never a moment when a jaw had to stop working so an ear could start.

The fossils that show this exist. And the connection was worked out from embryos *before* anyone found the fossils — which then confirmed it.

Tracks in you

Early embryos of fish, birds, and humans all develop the same set of ridges in the neck region. In fish they become parts of the gill support. In you they become your jaw, parts of your middle ear, and your voice box.

The starting material is shared. What it turns into is not.

(To be clear, because there's an old wrong version of this: human embryos do **not** have gills, and an embryo

does not replay its evolutionary history. What's shared is the raw material, not the sequence.)

Some leftovers are simpler. Wisdom teeth often don't fit, because the jaw they evolved for was longer than the one you have. Your tailbone is a set of fused vertebrae from a tail you don't have — though it isn't useless, since muscles attach to it.

Put together

All of this is the signature of a process that can only edit what's already there.

It can't set something aside and start over, because every single intermediate stage has to be a working, living animal. So it reaches its answers by whatever route was available rather than the route that would be shortest — **and the route stays visible in the result.**

A giraffe's voice box is wired through its chest because a fish's was wired through where its heart used to be.

10
Spirals

Once you start looking, spirals are everywhere, and some of the best ones are lying around the house: a garden hose coiled for the winter, a rope looped over a hook, the line on a fishing reel. Keep the hose in mind. It's doing the same job as something inside most of your cells, and this chapter comes back to it.

Then go outside. Snail shells and ram horns spiral. A sunflower head and a pine cone are built from interlocking spiral rows — count them sometime. A pea tendril coils around whatever it touches. Bathwater leaves in a whirlpool, and hurricanes on a weather map turn. Some galaxies have spiral arms, and DNA is a twisted ladder.

The usual reactions are "coincidence" or "it must mean something deep." Both are wrong, and why is more interesting than either.

Four ways to make a spiral

A spiral isn't one thing with one cause. Several unrelated processes produce it. The pattern is real; the explanation is plural.

Grow at the edge, in proportion. A snail adds shell only at the opening, each new section bigger than the last by a fixed proportion. Do that, and the shape *can't* be anything but a spiral. The snail isn't drawing a spiral. A spiral is what steady growth at an edge looks like from outside.

Turn while moving forward. Anything that rotates as it advances traces a helix — a corkscrew, a drill bit, the twisted ladder of DNA. A tendril coils because one side ends up shorter than the other. Same shape, unrelated causes.

Pack things in one at a time. This one looks the most designed and is the least. A sunflower makes seeds one at a time at its centre, each in the biggest gap left. The angle between seeds settles at about 137.5 degrees, and the interlocking spirals follow. The proof it isn't about plants: in 1992, researchers dropped magnetized droplets into a dish where each new drop was pushed away by the ones already there. **They got the same spirals.** No genes, no plant. The pattern comes from the packing problem, not from the thing solving it.

Spin while flowing inward. Water turning as it drains speeds up as it nears the plughole, the way a skater spins faster by pulling her arms in. Whirlpools and hurricanes work the same way. (Your sink doesn't spin because of the Earth's rotation. The shape of the basin and the water's leftover motion decide the direction.)

What they have in common

A spiral is what growing looks like when something has to keep its connection to where it's already been. You can see it in a spider's web and in a spiral staircase.

The shell can't rebuild what's behind its lip. The corkscrew can't occupy the space it just left. The seed head can't push its old seeds out of the way. Each has to extend without undoing its own past, and very few shapes allow that. A spiral isn't nature's favourite. It's what geometry leaves available.

That makes a prediction you can check: things free to rearrange themselves, or already settled, shouldn't spiral. Mostly, they don't. A mountain is a heap holding itself up. A planet is a sphere. A salt crystal repeats one tiny cube. Those are shapes of **having arrived**. Spirals are shapes of **still going**.

My own idea. Not a fact, only a way of seeing it: a potter's wheel lets a person work *with* a spiral instead of drawing one. The wheel does the turning. The potter's hands do the moving, drawing the clay up and out as it spins, and turning while moving forward is how you get a helix. The pot is what that path leaves behind.

Spirals people make

People have been carving spirals for a very long time. Newgrange, a huge stone tomb in Ireland, was built around 3100 BCE, more than five thousand years ago. Many of its stones carry carvings that come in a few repeating families: curvy ones, like circles and spirals, and straight-edged ones, like zigzags, parallel lines and diamond shapes. The carving at the entrance puts spirals and diamonds side by side, and so does the stone directly opposite, at the back of the mound.

Deep inside, a triple spiral is carved on the wall of a room that the rising sun reaches once a year, on the shortest day of winter.

What any of it meant to the people who carved it is lost.

The garden hose problem

Back to the hose. Coil one by hand, loop after loop the same way, then pull it straight. It kinks. Every loop put a twist in the hose, and the twists have nowhere to go.

That's a rule, not bad luck: for a long strand whose ends can't spin freely, **the amount it twists plus the amount it coils stays the same.** People who handle long things for a living have found every way around it:

- **Turn the drum, not the hose.** A hose reel spins as you wind, so the hose never has to twist.

- **Cancel it out.** Stage crews coil cables "over and under," alternating the twist from loop to loop, so the cable lies flat when it's thrown out.
- **Let it escape.** A fishing reel whose spool stays still twists the line on every cast, so anglers tie in a swivel — a tiny joint that spins and lets the twist run out.
- **Put it to work.** Rope is fibres twisted one way, bundles twisted the other way, then those twisted the first way again. Each layer locks the next, which is why rope holds together at all. People have made twisted rope for at least 17,000 years.

Now your cells. Most kinds of cell in your body hold 46 very long strands of DNA, wound around tiny protein spools — about thirty million in each cell — and folded again. A strand coiled on top of being twisted is called **supercoiled**. And DNA has to be read, in order, like line coming off a reel. Opening it up to read it piles twist up ahead and leaves too little behind: the fishing reel's problem. Cells solve it with proteins that cut the DNA, let it spin free, and join it back up. They're the swivel.

Settled. The twist-and-coil rule is real mathematics, partly worked out *because* of DNA. The hose, the rope, and the chromosome all obey it.

My own idea. A coil is the shape of something long that's meant to be paid out later, in order. That's this chapter's idea again: extending without undoing your past. Whether it's more than a good way of seeing it is the kind of question the last chapter asks.

So the next time a hose kinks on you, you're looking at a problem your cells are solving all day long.

Two warnings

Looking alike is not working alike. A snail shell and a spiral galaxy look similar in photos and have *nothing* in common underneath. Galaxy arms aren't material winding up over time. They're waves of compression moving through the disc, with stars drifting in and out of them.

"Spiral" is doing loose work. A shell, a helix, a sunflower head, and a whirlpool are different objects. One word for all of them is convenient, and convenience is where arguments go wrong.

Where I'd push it, and why you shouldn't follow

> **My own idea.** Things sort into two groups: settled ones, shaped by the forces on them right now, and running ones — growing, circulating — whose shape also records the path they took. The spirals are all in the second group.

Now how that goes wrong, with me as the example. There's a spiral of hair at the crown of your head, right above the part of the brain most tied to deliberate thought. It's very tempting to make something of that.

Don't. Hair whorls come from the direction follicles point as skin develops, and have nothing to do with the brain underneath. The "connection" exists only because both are on top of your head. I feel its pull every time I think about this chapter — exactly the pull chapter 13 is about. Noticing it and refusing it is the only defence there is.

11
Time that stacks up

Lots of things in nature repeat. Day and night. Seasons. Tides. Your heartbeat. Orbits.

But a cycle almost never comes back to exactly where it started, because the thing doing the cycling has been changed by the previous rounds. You get repetition *plus* a residue. Round and round, and also forward.

Cycles that keep their own records

This isn't abstract. Several natural systems write the record down, which is most of how we know anything about the deep past.

A tree adds a ring every year. The cycle is the same each time; the ring isn't. Its width records that year's rain, temperature, and crowding. And because rings stack instead of overwriting, a trunk holds the years in order. By matching patterns across living trees, dead timber, and wood from old buildings, that record has been chained back thousands of years.

Ice does it with snow. Each year's snowfall gets squashed by the next, and bubbles of air get sealed in as it closes over. The longest Antarctic record goes back

about **800,000 years**, through eight ice ages. And here's the part worth sitting with: the ancient air is still in there. The carbon dioxide levels of 800,000 years ago aren't estimated from something else. They're *measured*, out of the actual air.

Lake beds layer by season. Some corals band with the tides.

In every case, a cycle that could have just repeated instead left a residue — and the residue is what makes history readable at all.

There's a name for a system whose current state depends on the path it took, not just on where it is now: **hysteresis**. It comes from magnets, which remember what was done to them. The same structure shows up anywhere a cycle leaves something behind.

The same shape in bigger things

Evolution works this way. Each generation starts from whatever the last one produced.

So does learning. Practising something changes what practising it does next time.

So does growing up. Everyone goes through roughly the same stages, and nobody goes through them identically, because each stage builds on how the one before it actually went for that person.

So does culture. Practices get passed on, and what gets passed on has been reshaped by every previous passing.

None of those is literally a spiral, and the word showing up this often should make you suspicious rather than satisfied. What they genuinely share is structural: **something that repeats, inside something that keeps the consequences of its earlier repeats.**

The last chapter described this in space — extension that can't go back over itself. This is the same constraint in time. A cycle that left nothing behind would be a closed loop, and a world made of closed loops would have no history in it to read.

12
What counts as alive

You know the list from school. Living things use energy, grow, respond to their surroundings, reproduce, pass on traits, and evolve.

The list works for most cases. It falls apart at the edges, and how it falls apart is more interesting than the list.

Things that do some of it

Fire uses energy. It consumes fuel, releases heat, keeps itself going while fuel lasts, grows, responds to its surroundings, and spreads from place to place in a way that looks a lot like reproduction.

What fire has none of is **heredity**. One fire doesn't pass traits to the next. Nothing accumulates, so nothing ever improves. Fire is the cleanest proof that the energy-and-growth half of the list isn't enough on its own.

Crystals grow by adding material in an ordered pattern, and they copy structure — drop a seed crystal into the right solution and you get more of the same form. Pattern really is being copied. But there's almost no

room for variation that could be inherited and selected, so again, nothing builds up.

Prions are proteins that spread by making other copies of the same protein fold wrong, the way they're folded. Structural information passes from one molecule to another with no DNA or RNA involved at all. Whether that counts as heredity depends entirely on how you define the word.

Viruses carry genes, mutate, and evolve. They are absolutely subject to everything this book describes. They also can't eat, can't make energy, and can't reproduce without hijacking a living cell. People have argued about whether they're alive for a century.

And that argument has got *harder*, not easier. Giant viruses found in the last twenty years have more genes than some free-living bacteria, and carry bits of protein-building machinery that viruses were supposedly defined by not having. Meanwhile some bacteria — including the ones causing typhus — have lost so much of their own metabolism that they can't reproduce outside a host either.

The property that was supposed to separate viruses from organisms turns out to belong cleanly to neither.

Why no definition has fixed this

There's no agreed definition of life, and the disagreement isn't close to being resolved. What follows is why.

People have tried. One survey counted **over a hundred and twenty** published definitions of life.

That number is the most useful fact in this chapter. A thing with one definition has been pinned down. A thing with a hundred and twenty has not.

The most-used attempt — written for NASA, who needed something practical for looking for life on other planets — is "a self-sustaining chemical system capable of Darwinian evolution." It's compact and it does real work.

It also has an obvious hole. A single organism that can't reproduce — because it's sterile, or old, or the last of its kind — isn't capable of Darwinian evolution by itself. Nobody would say it isn't alive. The definition describes populations better than individuals, and fixing that tends to let in things it was meant to keep out.

Every proposed definition does some version of this. It either rules out something that is plainly alive, or lets in something that plainly isn't. The usual fix is another clause, and enough clauses later you have a definition that's accurate and says nothing.

The thing to understand about that

Here's the part worth slowing down for, because the obvious reading is wrong.

Think about a heap of sand. Take away one grain — still a heap. Take away another — still a heap. You can

never point to the grain where it stopped being a heap. But one grain is definitely not a heap.

That's not because nobody has studied sand carefully enough. The word just has fuzzy edges. No amount of further research will sharpen them, because there's nothing there to sharpen.

"Life" looks like that kind of word. It picks out a bundle of properties that usually come together, and it works perfectly well in ordinary cases precisely *because* they usually come together. It gives no answer at the edges because it was never the sort of word that had one.

And that matters for everything in chapters 3 through 5, and chapter 8.

"When did life begin?" is a broken question, in the same way "when did it become a heap?" is broken.

Those chapters describe properties piling up one at a time — chemistry that lasts, boundaries that hold things in, networks that feed themselves, patterns that copy. Each step is small. Each step is chemistry. No step needs anything the one before it didn't have.

The question expects a moment. There isn't one. And that absence is not a gap in the account.

But don't let me off too easy

That is not the same as saying there's no problem.

There's a large one, and you already met it in chapter 8: nobody knows how a chemical system first came to copy a pattern reliably enough for differences to accumulate.

What dissolves is the expectation of a *single dramatic moment* needing a single dramatic explanation. What's left is a series of steps, at least one of which is genuinely unexplained.

That's a different shape of not-knowing. It's also a more useful one, because you can go looking for a step.

One more limit

Every property on that opening list came from studying examples of exactly one kind: life on Earth, carbon-based, water-dependent, made of cells, all descended from a common ancestor.

The definition of life is a generalization from a single case.

You can't know whether a category built from one example is general. There might be organized, energy-driven things out there that fail all our criteria and are still the kind of thing we were reaching for. There might be none. The possibility is open — and being open is not the same as being evidence.

13
What a brain throws away

Nervous systems let an animal take in signals, coordinate a body, and react. Bigger brains do more: they build *models* — internal versions of the outside world that can be used to remember, predict, and decide.

The surprising thing about that is how much gets thrown out to do it.

Ninety-nine per cent, before you even see it

Count the light sensors in a human eye. There are about **ninety-six million** of them in one retina.

Now count the wires leaving the eye for the brain. The optic nerve has about **one million** fibres.

So roughly a hundred sensors feed each outgoing wire. Your eye discards about ninety-nine per cent of what it collects **before sending anything.** (It isn't even — at the centre of your vision each sensor gets at least one wire to itself, maybe two, and at the edges one wire serves hundreds. Which is why you can read this sentence but can't read one in the corner of your eye.)

And what does get sent isn't a picture. It's already been processed into edges, movement, contrast, and

change. Your eye isn't a camera. It's a device that extracts a few useful things and throws the rest away.

Then it happens again, several times, before anything reaches your awareness.

So seeing a room doesn't mean holding a room in your head. It means holding a model that's good enough to act in the room.

Every mental thing on the list discards something

Your brain builds that model out of regularities. It notices what recurs, treats recurrences as examples of a category, and keeps the category instead of the examples.

That's why you can recognize a dog from an angle you've never seen. It's also why eyewitnesses drift towards describing a *typical* version of what they saw.

Look at what each mental process costs:

- **Attention** throws away almost all of what's available.
- **Categories** throw away the differences between things in the category.
- **Memory** throws away detail and rebuilds it on demand — which is why remembering is more like reconstructing than like replaying.

What you buy with all that loss is **prediction**. A system carrying a compressed model of its world can guess

what's about to happen and move first. An animal that only reacts to what's in front of it is always late.

Notice this is the same deal as chapter 8. The genetic code tolerates mistakes where mistakes are cheap and resists them where they're expensive. A brain throws away detail that wouldn't change what to do and keeps detail that would.

Neither was designed to compress. Both ended up compressing, because **a system that keeps everything can't act in time.**

A model that includes you

A model of the world is useful. A model that includes the thing doing the modelling is a different kind of object.

An animal that represents its own body, its own position, and eventually its own representing has started doing something that loops back on itself. Douglas Hofstadter called this a **strange loop**: go down through the levels of a system and you come out back at the top.

His argument is that the feeling of being a single unified "you" is what that loop feels like from the inside. Not a thing sitting somewhere in your head. A pattern your brain makes by modelling itself modelling.

Marvin Minsky got to the same place from the other direction. On his account your mind was never unified to start with — it's a large crowd of specialized processes handling vision, sound, balance, memory, plan-

ning, fear, none of which contains a "you," which between them produce the appearance of one.

A flock of starlings is a useful comparison, as long as you hold it as a comparison. A flock turns like a single thing and looks directed. No bird is directing it. Each one reacts to a few neighbours, and the shape is what all those local reactions add up to. **The pattern is completely real. The thing deciding it is not.**

Starlings and neurons have almost nothing in common. But the comparison makes the point that something can look unified without anything unifying it.

What this does not explain

One thing has to be said plainly.

None of that explains why any of it is *experienced*. Why there is something it is like to be you, rather than all this processing happening in the dark.

Hofstadter and Minsky describe an architecture. Describing an architecture does not answer that question, and this book does not answer it either. It may not be the kind of question that gets answered by describing machinery more carefully.

The thing that goes wrong

These abilities are powerful and they are not reliable, and this is the most practical paragraph in the book.

The faculty that finds real patterns also finds patterns that aren't there. Humans are unusually good at noti-

cing that two unrelated things seem to work alike — which gives us analogy, invention, science, and storytelling. It also gives us confident connections between things with nothing to do with each other.

And there's no reliable internal signal telling you which is which. In experiments, answers that arrive with an "aha!" are right more often than answers worked out step by step, and the "aha!" is stronger when they're right — but wrong answers can set it off too, and an "aha!" from solving something unrelated can make a false statement feel truer.

That's the problem the opening of this book describes from the inside. It's why every claim here is tagged. The tags aren't modesty — they're the only available correction for a faculty that can't reliably check itself.

14
Words

Language is small pieces combining into bigger ones. Sounds into words, words into sentences, with rules about what can attach to what.

And it changes the way everything else in this book changes. New words appear, some catch on, most don't. Pronunciations drift. Meanings move. Nobody is in charge of it.

Some sounds recur across unrelated languages for a plain physical reason. Words like *mama* and *dada* turn up all over the world partly because those are among the easiest sounds for a baby's mouth to make reliably — and there's usually a parent nearby, happy to assume it's about them. Similar words can come from a shared constraint just as easily as from a shared ancestor.

Does your language change how you think?
This is where popular writing and actual evidence come apart worst, so it's worth doing slowly.

The famous example, and why I'm not using it

You've probably heard that Inuit languages have an enormous number of words for snow, and that this lets their speakers see distinctions the rest of us can't.

Watch how that claim grew.

In 1911, an anthropologist noted **four** root words — snow on the ground, falling snow, drifting snow, and a snowdrift. He didn't suggest four was a lot or that it meant anything. A later writer repeated it with **seven**, three of which he didn't source. Later retellings said dozens. Then hundreds. Each one citing the retelling before it, nobody going back to check.

A linguist eventually traced the whole chain, and the story is now a standard example of how a factoid grows.

So I'm not using it. But there are two things underneath it that are true, and leaving them out would be its own kind of dishonesty.

First, counting words is the wrong question for those languages. They build long words by stacking pieces together, so the number of possible "words" for anything is effectively unlimited. A count doesn't measure anything.

Second — and this is the part the debunking usually skips — when researchers actually went and asked, the vocabularies were real. Surveys across about ten Inuit and Yupik dialects recorded something like seventy terms for ice conditions in one

Alaskan dialect, around forty snow terms in another language, about fifty in a third. These are not synonyms. They mark whether a surface will hold your weight, how it formed, and how it's likely to break.

So the popular version is wrong in its *form*, and there's something real underneath it. Both halves have to be said.

What a landscape does to words

When I started on this chapter I expected the snow story to have cousins. Forest peoples, I assumed, would have extra words for shades of green. Desert peoples would have extra words for the colours of sand. Checking showed that isn't what happened — and what did happen is better.

Forests don't make more words for green. Researchers who tested colour words with the Tsimane', who live in the Amazon rainforest, found they use colour words *less* than people in cities do, not more. The reason seems to be what a colour word is for. Across more than a hundred languages, colours get named most precisely when they belong to *things* — a ripe fruit, a red flower — and those are mostly warm colours. Green leaves and blue sky are the background. You rarely need a word to pick out the background.

The Hanunóo, farmers in the hills of the Philippines, do have a word that covers green. But it also means *fresh*. A newly cut piece of bamboo gets the "green"

word even though it's brown, because it's fresh and wet. Their word that covers red also means *dried out*. The words track what matters about a plant — fresh, or past it — the way Inuit ice words track whether ice will hold you.

Deserts don't make more words for the colour of sand. But they make words for the *ground* — and those words were useful enough that science borrowed them. Geologists describing deserts say *erg* for a sea of dunes, *hamada* for a bare rocky plateau, *sabkha* for a salt flat, and *wadi* for a riverbed that's dry until a storm fills it. All four are Arabic. A crescent-shaped dune is a *barchan*, from Kazakh. The cold has done the same: *nunatak*, a peak poking up through an ice sheet, is Greenlandic; *pingo*, a hill with a core of ice, is from Inuvialuktun, spoken in Arctic Canada.

So the pattern is real, but it isn't about colour. **A landscape fills a language with words for the differences that can hurt you or feed you** — firm ground or soft, fresh plant or dried, safe ice or thin. Colour words follow something else: what has to stand out against the background, and how much people use colour to tell things apart. Places full of dyed and painted objects tend to need more of them.

Still argued about. Why some languages have more colour words than others. Usefulness is one strong explanation. Another links places with the harshest sunlight to having no separate word for blue — but when researchers tested the reason given for it, the test failed. Probably several things matter at once.

What the actual experiments found

These studies test what people *do* instead of counting their words, which makes them better evidence. It also makes it possible to see when a result doesn't hold up, and one of these didn't.

Colour. Russian has no single word covering what English calls blue — it has separate everyday words for lighter blue and darker blue, and you have to pick one. In a 2007 experiment, Russian speakers were faster at telling two blues apart when the pair crossed that boundary than when both were on one side. English speakers showed no difference.

Then the clever part. The experiment was repeated while people held a string of digits in memory, and again while they held a shape in memory. The Russian advantage **disappeared under the word-based load** and survived the shape-based one.

And then a 2020 attempt to repeat that study didn't find the effect at the boundary the whole claim rests on.

I'm telling you that because this is the study this section was built on. It would be easy to leave it out and nobody would notice.

Still argued about. What the colour result means. The original reading was that the language was being used live, during the task, rather than having permanently rewired anyone's eyes. After the failed repeat, the honest position is smaller: there's evidence that the words you use habitually get recruited while you think, the colour case is the most famous demonstration of it, and the colour case has a serious failed replication.

Number. Some languages have very few counting words. The most discussed is an Amazonian language two of whose quantity words were first translated as "one" and "two," and later reinterpreted by the same researcher as closer to "a small amount" and "a larger amount" — and an experiment backed him up: speakers used the supposed word for "one" for as many as six things.

What the experiments found is specific and worth getting right. Speakers matched exact quantities accurately, **including large ones, when the objects were in front of them.** Accuracy fell apart when the objects were shown and then hidden before they had to reproduce the amount.

Still argued about. What that means. The difficulty doesn't seem to be seeing or matching exact quantity. It's **holding one in your head without a symbol for it.** That's a much narrower claim than "they can't count," and a far more interesting one: number words look like a tool for memory rather than a tool for perception.

What to take from all this

The strong version — that your language makes some thoughts impossible — is not supported and isn't claimed here. People notice distinctions their language doesn't have, borrow words, invent words, and talk around things. Nobody is trapped.

Still argued about. The narrow version: words you use constantly get recruited while you do mental work, and they affect how fast and how accurately you do it. Where the effects show up at all they're small, and replication in this field is genuinely patchy.

And the influence runs both ways, which makes "which caused which" usually the wrong question. A community that needs to tell ice conditions apart develops words for them — and then the words make the distinctions easier to notice and teach. **Vocabulary is as**

much a record of what has mattered to people as a cause of what they notice.

What a word costs

A word is an extreme act of throwing things away. One short sound stands in for a whole category of things, no two of which are identical.

The saving is enormous. The loss is real, and it's the same loss as in chapters 8 and 13. Every category discards the differences between the things inside it. Naming something makes it easier to think *with* and slightly harder to see freshly, because the name arrives with expectations attached.

The structure of a language is a record of which distinctions people found worth the trouble of keeping — and, by what's missing, which ones they didn't.

Why it matters most

Here's the consequence that outweighs the rest.

Language lets something formed in one mind get rebuilt in another. And rebuilt again after the first mind is gone.

Before that, everything an animal figured out died with it, and the next generation started over. After it, knowledge piles up. A skill can be handed on, improved, and handed on again — and **nobody in the chain has to understand why it works.**

Which is the subject of the next chapter, and it turns out to be more consequential than it sounds.

15
Knowing without knowing why

Culture is everything people pass to each other — skills, beliefs, tools, stories, institutions. It varies, it spreads, some of it sticks, most of it doesn't.

That sounds like evolution, and it rhymes with it, with two differences big enough to matter. Culture moves far faster, and it doesn't only go from parent to child — it jumps between strangers, across countries, and backwards in age from a kid to a grandparent. It's also copied *badly*. A gene is copied as a sequence. A skill is rebuilt from watching, by someone who may have misunderstood it.

There's a word for one of those copied bits. In 1976 Richard Dawkins, a biologist who studies evolution, wrote a book called *The Selfish Gene*. In it he made up a word for a tune, a catchphrase or a fashion that spreads by being copied from one person to the next, the way a gene is copied from parent to child. He called it a **meme**. He took *mimeme*, from an old Greek word meaning "something imitated", and cut it short so it would sound like *gene*. Today the word mostly means a

funny picture that spreads online. That's a newer, narrower use of the same word.

> **Still argued about.** Whether memes really explain how culture works is an open question. Ideas aren't neat, separate pieces the way genes are. They get copied far less faithfully, and they can't be pulled apart from the people who hold them. The word is handy. Whether it describes what's actually happening is another matter.

That sloppy copying turns out to be the interesting part.

The most important thing in this chapter
A practice can carry real knowledge that nobody performing it has.

Here is the clearest example I know.

Corn

Maize was domesticated in Mesoamerica thousands of years ago and became the staple crop. And from early on, people prepared it a specific way before eating: soak and cook the kernels in water mixed with wood ash or lime, then hull and grind them.

The reasons to do this are obvious if you're the one cooking. The tough skins come off easily. The ground corn holds together into dough instead of crumbling. It smells and tastes better.

The reason that actually mattered was invisible.

Maize is low in niacin — vitamin B3 — and low in tryptophan, which your body converts into niacin. Live on untreated maize as your main food and you don't get enough of either.

Niacin deficiency is a disease called **pellagra**. It causes skin lesions, digestive collapse, neurological damage, and eventually death.

The alkaline soak prevents it. It makes the corn's protein more digestible, which makes its tryptophan available, and your body makes its own niacin from there.

And here is something worth stopping for.

The explanation you'll usually find — that corn contains locked-up niacin and the alkali releases it — is wrong. Multiple experiments have failed to support it. It is also the explanation *I* believed when I started writing this chapter, and the paragraph above had to wait until it was checked.

So look at the shape of that. The practice worked for more than a thousand years, and maybe for over three thousand, with nobody knowing why. Then it was explained. Then the explanation turned out to be wrong. And the wrong explanation is still the one in general circulation, including in my own head until recently.

Knowledge hidden inside a practice is hard to dig out. Knowledge hidden inside a *plausible story* about a

practice may be harder, because a plausible story stops people looking.

Then corn got on a boat

This is where it becomes an experiment nobody designed.

Maize spread fast through Europe, Africa, and the American South, because it grows where other grains won't and produces a lot of calories per acre.

The cooking technique did not travel with it.

And why would it? It looked like a regional culinary preference. Nobody on either side of the exchange knew it was doing anything.

Pellagra followed the crop. Outbreaks in northern Italy, Spain, southern France, Egypt. In the American South it became a mass health crisis: between 1906 and 1940 it affected **more than three million people and killed over a hundred thousand.** At its worst in the late 1920s it was among the leading causes of death in the region.

For years it was blamed on an infection. Often it was blamed on the character and habits of the people getting sick — who were overwhelmingly poor, and whose poverty was the actual reason their diet had narrowed to cornmeal in the first place.

It took deliberate investigation in the 1910s and 20s to establish it was dietary, against serious resistance. The specific vitamin wasn't identified until 1937.

Mesoamerican cooks had solved the problem centuries or millennia before anyone could state what the problem was — and could not have stated it themselves.

What that means

Sit with it for a second. The knowledge was real. It was accurate. People's lives depended on it. **And it did not exist in anybody's head.**

It existed in a sequence of actions, passed from one person to the next, with no explanation attached beyond *this is how you make corn.*

A practice is a compressed version of a relationship with the world — and it's lossy in a specific, dangerous way. **It throws away the reason.**

That makes it cheap to transmit: a child can learn it by watching and needs to understand nothing. It also makes it fragile, because a practice separated from its reason can't defend itself. Anyone can see a step is tedious. Nobody can see what it's for.

And it's a correction to a common bit of condescension. "They didn't know why" gets heard as "they didn't know." The corn case shows those are different claims, and the second one can be false while the first is true.

Religion, carefully

Religion can be looked at the same way — as a system of explanations, stories, rituals, and institutions that get passed between people and across generations.

I want to be clear about what this section does. It looks at **how** such systems are organized and transmitted. It does not assess whether any of them is true, and nothing here should be read that way.

People have always faced things that were urgent, arrived without warning, and could not be investigated: death, illness, weather, famine, disaster, where everything came from. Shared stories do real work under those conditions. They make events interpretable. They give a community a common frame to coordinate around. They carry moral expectations in a form people remember. They offer a way of facing death that can be held together rather than alone.

Here's one specific experimental finding, because it's narrow and checkable.

When you pass a story along a chain of people — each one retelling it to the next — content that's **mildly** strange survives better than content that's either completely ordinary or completely bizarre. A story where almost everything behaves normally and one thing doesn't gets retold more faithfully than one where nothing is odd, or where everything is.

That has an obvious bearing on the shape religious stories tend to take across completely unrelated traditions: a mostly ordinary world containing a few specific exceptions.

It says nothing at all about whether any of them is true. It says something about which stories, true or false, are likely to still be around.

And that distinction needs stating plainly, because it's easy to misread. **Finding a function is not an explanation of origin and it is not a debunking.** That something is useful is compatible with it being true, with it being false, and with the question being the kind that doesn't get settled this way. Plenty of true things are also useful. Usefulness is evidence of neither.

The functional question and the truth question are different questions. Running them together is a mistake in either direction.

16
What we don't know

Here is the whole book in six lines.

Chemistry finds ways to last. Life finds ways to copy. Bodies find ways to split up the work. Brains find ways to model. Language finds ways to transmit. Culture finds ways to pile up.

Or, as Ian Malcolm, Jeff Goldblum's chaos theorist in Spielberg's *Jurassic Park* (1993), puts it: "Life, uh, finds a way."

Every one of those sentences is slightly dishonest, and it's better to say so at the end than to let the rhythm carry it past you. **None of those things *finds* anything.** Nothing is searching and nothing is wanted. The phrasing is a compression — exactly the kind this book has been describing — and like all compression it throws something away. What it throws away here is the absence of anybody doing it.

Where the account holds, and where it doesn't

The honest summary is that the continuity is real and uneven.

Solid in principle, from physics through bags. Chemistry is physics. Self-sustaining reactions are chemistry: chapter 4 shows them working, and small ones built in labs. Whether early Earth's chemistry was rich enough to make them on its own is still argued about, as chapter 4 says. Bags form on their own, and the competition between them in chapter 5 isn't *like* natural selection — it is differential survival, happening to bags, watched in a dish. Nothing new is needed at any step. What's open is whether early Earth had the right conditions, not whether the chemistry can happen.

A real break at copying. Nobody knows how it started. Chapter 8 says so and this chapter says so again.

Solid again, from copying through brains. Once things copy with variation, ordinary biology carries you through multicellular life, specialization, nervous systems, and animals that model their world. This stretch is the best-supported part of the book and the least original to it.

Shakier at language, and this book says little about it that a linguist would respect.

Above language, it's a shared vocabulary and not much more. Cultural selection and biological selection get described with the same words, and the machinery is unrelated. That's chapter 2's warning, and it applies hardest at the top.

The strongest objection to all of this

Here is the serious objection to this book, put as well as I can put it:

> It's an interesting story and too broad to count as a real theory. It links a lot of fields by analogy without isolating sharp, testable mechanisms that would let it be decisively right or wrong. It risks explaining too much with one pattern. And it doesn't predict anything that ordinary science in those fields wouldn't predict anyway.

Every clause of that is fair.

The worst one is "explains too much." **A pattern you can find everywhere constrains nothing.** A framework that no observation could embarrass isn't telling you anything.

And it gets worse. The ability to find patterns everywhere is exactly the unreliable faculty from chapter 13 — and exactly what the opening of this book describes me doing at unusual intensity. The person most likely to see this pattern in everything is the person already primed to see it. Being careful doesn't fully fix that.

Checked, and it doesn't help as much as it looks

An earlier version of this chapter listed three claims this framework makes that could be tested, and presented them as experiments somebody should run.

Then came the check: **all three had already been run.** Cycling really does help prebiotic chemistry accumulate. Enclosing things in bags really does help them persist and adapt. Cultural transmission really does have measurable biases in what survives.

All three came out in the book's favour. That's a worse result than it sounds, and here's why.

Those predictions weren't derived from the framework and then tested. They were assembled by someone who'd read around these fields, pointing in the directions the evidence already pointed. **Passing a test you picked after you knew the answer proves nothing.**

What would count is a prediction this framework makes that a specialist wouldn't expect. I don't have one. Until I do, the objection above stands.

The four things nobody knows

How copying started. The biggest one. Named several times already, and not narrowed by anything in this book.

Why any of it is experienced. Chapter 13 describes machinery. It doesn't explain why there's something it's like to be you, and it may not be the kind of question machinery answers.

Writing about this gets strange fast. To describe consciousness, I have to use my own brain to describe that same brain, while it's busy doing the describing. Putting

that into one sentence took me more than ten tries. It's worth reading twice.

Whether complexity actually goes anywhere. This book describes organization piling up. It also says evolution has no push towards complexity. Both are defensible and I haven't squared them.

> **Still argued about.** Whether anything pushes life towards complexity at all.

Here's the uncomfortable alternative. Imagine a wall, and something wandering randomly next to it. It can't go through the wall, so over time it drifts away from it — **with nothing pushing it.** If there's a floor below which life can't get any simpler and still work, then random variation alone will spread things out away from that floor, and you'd see exactly what we see.

Where this has been studied properly, the evidence has tended to favour that boring explanation over a drive towards complexity, and so against the version of the story this book is built around. The most common organisms on Earth are still single-celled. What's increased is the *maximum*, not the typical.

A book organized around things getting more complicated has an obvious reason to underweight that, and probably has.

Whether groups do what minds do.

My own idea. A shaky one. Individual minds model themselves. Cultures accumulate models. Whether a whole society can be said to model its own modelling in any way that isn't merely a metaphor is unclear, and people treating it as settled in either direction haven't convinced me.

What would prove this wrong

Fair question, and the answer is thinner than I'd like.

The cheapest test is one nobody has run, including me. **Generate systems with no real structure in them, describe them in this book's vocabulary, and see whether anyone can tell them from the real ones.** If the words fit random junk just as well, then this book describes the describer and not the world.

Beyond that: if someone found a single mechanism that genuinely did unify all these fields, the book's central hedge would be wrong in the opposite direction. And if the origin of copying gets explained by chemistry that owes nothing to variation, selection, or compression, then this vocabulary was decoration over the interesting part.

That last one is the most likely of the three.

One more honest note

There's a spare organ I was going to use in chapter 9 as an example of a useless leftover: the appendix. I cut it,

because the evidence has shifted and it now looks like it has immune functions and acts as a reservoir for gut bacteria.

Worth keeping the reason. **"We don't know what this is for" and "this has no purpose" are different claims**, and the first keeps getting mistaken for the second. That's a good habit to carry out of this book and into everything else.

So what is it all for

The point of the vocabulary is to make a set of changes easier to see and talk about, without pretending they work the same way. That's a modest aim and it's the one the evidence supports.

You can finish this book believing very different things about what it adds up to. Nothing here is built to move you. It doesn't settle whether there's a purpose behind any of it, whether the patterns point at anything, or what to make of the fact that some matter on one planet ended up able to ask.

What it offers is an account where the steps connect. Chemistry that lasts, boundaries that hold, networks that feed themselves, patterns that copy, systems that model, minds that tell each other what they modelled.

One word there needs care. There *is* a break in this account, at copying, and it's not small.

But there are two meanings of "break" and they have to be kept apart. A break in the **account** is a hole in

what we understand, and this book points at its holes. A break in the **sequence** would be a place where something showed up that the previous step couldn't supply — and in every transition that has been worked out in detail, nothing like that has been found.

That's a claim about the steps we understand. It is not a promise about the one we don't.

Which is where the book ends, and where it said it would. The questions it leaves open are left open on the page, with the arguments still standing next to each other, because that's more useful to you than a tidy ending would be.

The way to tell a real pattern from a convincing one is to go and check.

Where to look next

Everything in this book came from somewhere, and most of those somewheres are more interesting than the summary.

If the chemistry grabbed you, look up the **formose reaction**, **Bénard cells**, and the **Belousov–Zhabotinsky reaction** — the last one you can watch on video, and you should.

If the bags did, search for **protocells** and **fatty acid vesicles**.

If evolution's leftovers did, look up the **recurrent laryngeal nerve** and the **evolution of mammalian**

ear bones, which is one of the best-documented transitions in all of palaeontology.

If the spirals did, look up **phyllotaxis** and the **Douady and Couder** droplet experiment.

If brains did, read Hofstadter's *Gödel, Escher, Bach* — it's long, it's strange, and it's worth it.

If the corn did, look up **nixtamalization** and **pellagra**, and notice how recently this was all worked out.

And if you want to know how reliable any particular claim in this book is: chapter 17, at the back, names the sources and marks the ones whose support hasn't been fully checked yet. The longer edition also says which chapter leans on each one.

17
Where this comes from

These are the sources behind this book. It's the same list the longer edition uses, because both editions rest on the same research. The longer edition also says which of its chapters lean on each source.

Every entry was checked against the publication record for its authors, year, title and other details. Whether each source really supports the sentence that leans on it is a harder question. Most were checked against the source's own summary, and the most important claims much more deeply. A † means the sentence hasn't yet been checked against the source itself, not even its summary. Treat it as a pointer to where to look, not as proof that the looking is done.

Most entries end with a web address starting with https://doi.org/. That's a permanent link to the work.

References

Author–date, alphabetical: the same list as the longer edition's, without its notes on which of its chapters use each entry. Titles appear as their publishers print them. † marks an entry whose details are confirmed but

whose support for the sentence citing it hasn't yet been checked against the work itself.

Aktipis, C. A., Boddy, A. M., Jansen, G., et al. (2015). Cancer across the tree of life: cooperation and cheating in multicellularity. *Philosophical Transactions of the Royal Society B: Biological Sciences* 370(1673): 20140219. https://doi.org/10.1098/rstb.2014.0219

Ameta, S., Matsubara, Y. J., Chakraborty, N., Krishna, S., & Thutupalli, S. (2021). Self-Reproduction and Darwinian Evolution in Autocatalytic Chemical Reaction Systems. *Life* 11(4): 308. https://doi.org/10.3390/life11040308

An, D., Meissner, K. A., Nurowski, P., & Penrose, R. (2020). Apparent evidence for Hawking points in the CMB Sky. *Monthly Notices of the Royal Astronomical Society* 495(3): 3403–3408. https://doi.org/10.1093/mnras/staa1343

Anderson, S., Bankier, A. T., Barrell, B. G., et al. (1981). Sequence and organization of the human mitochondrial genome. *Nature* 290(5806): 457–465. https://doi.org/10.1038/290457a0

Andersson, S. G. E., Zomorodipour, A., Andersson, J. O., et al. (1998). The genome sequence of *Rickettsia prowazekii* and the origin of mitochondria. *Nature* 396(6707): 133–140. https://doi.org/10.1038/24094

Anthwal, N., Joshi, L., & Tucker, A. S. (2013). Evolution of the mammalian middle ear and jaw: adaptations and novel structures. *Journal of Anatomy* 222(1): 147–160. https://doi.org/10.1111/j.1469-7580.2012.01526.x

Aragón, J. L., Naumis, G. G., Bai, M., Torres, M., & Maini, P. K. (2008). Turbulent Luminance in Impassioned van Gogh Paintings. *Journal of Mathematical Imaging and Vision* 30(3): 275–283. https://doi.org/10.1007/s10851-007-0055-0

Archetti, M. (2004). Codon Usage Bias and Mutation Constraints Reduce the Level of Error Minimization of the Genetic Code.

Journal of Molecular Evolution 59(2): 258–266. https://doi.org/10.1007/s00239-004-2620-0

Aroskay, A., Martin, E., Bekki, S., et al. (2024). Geological evidence of extensive N-fixation by volcanic lightning during very large explosive eruptions. *Proceedings of the National Academy of Sciences* 121(7): e2309131121. https://doi.org/10.1073/pnas.2309131121

Ashtekar, A., Pawlowski, T., & Singh, P. (2006). Quantum Nature of the Big Bang. *Physical Review Letters* 96(14): 141301. https://doi.org/10.1103/physrevlett.96.141301

Balser, M., & Wagner, C. A. (1960). Observations of Earth–Ionosphere Cavity Resonances. *Nature* 188(4751): 638–641. https://doi.org/10.1038/188638a0

Barrett, J., & Nyhof, M. (2001). Spreading Non-natural Concepts: The Role of Intuitive Conceptual Structures in Memory and Transmission of Cultural Materials. *Journal of Cognition and Culture* 1(1): 69–100. https://doi.org/10.1163/15685370130006358 9

Bartlett, F. C. (1932). *Remembering*. Cambridge University Press.

Benner, S. A. (2010). Defining Life. *Astrobiology* 10(10): 1021–1030. https://doi.org/10.1089/ast.2010.0524

Boas, F. (1911). *Handbook of American Indian languages*, Part 1 (Bureau of American Ethnology Bulletin 40). Government Printing Office.

Boime, A. (1984). Van Gogh's *Starry Night*: A history of matter and a matter of history. *Arts Magazine* 59(4): 86–103. †

Bollet, A. J. (1992). Politics and pellagra: The epidemic of pellagra in the U.S. in the early twentieth century. *Yale Journal of Biology and Medicine* 65(3): 211–221. PMID 1285449.

Bollinger, R. R., Barbas, A. S., Bush, E. L., Lin, S. S., & Parker, W. (2007). Biofilms in the large bowel suggest an apparent function of the human vermiform appendix. *Journal of Theoretical Biology* 249(4): 826–831. https://doi.org/10.1016/j.jtbi.2007.08.032

Boroditsky, L. (2011). How Language Shapes Thought. *Scientific American* 304(2): 62–65. https://doi.org/10.1038/scientificamerican0211-62

Bourgault, D., Chavanne, C., Lemelin, J., et al. (2025). If van Gogh's *The Starry Night* Depicts Perfect Turbulence, So Should Degas' *A Woman Seated beside a Vase of Flowers. Bulletin of the American Meteorological Society* 106(8): E1703–E1707. https://doi.org/10.1175/bams-d-25-0004.1

Boyd, R., & Richerson, P. J. (1985). *Culture and the evolutionary process*. University of Chicago Press.

Breslow, R. (1959). On the mechanism of the formose reaction. *Tetrahedron Letters* 1(21): 22–26. https://doi.org/10.1016/s0040-4039(01)99487-0

Bressloff, P. C., Cowan, J. D., Golubitsky, M., Thomas, P. J., & Wiener, M. C. (2001). Geometric visual hallucinations, Euclidean symmetry and the functional architecture of striate cortex. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* 356(1407): 299–330. https://doi.org/10.1098/rstb.2000.0769

Brown, G. E., & Bethe, H. A. (1994). A Scenario for a Large Number of Low-Mass Black Holes in the Galaxy. *The Astrophysical Journal* 423: 659. https://doi.org/10.1086/173844

Burt, A., & Trivers, R. (2006). *Genes in conflict*. Belknap Press of Harvard University Press.

Butlerow, A. (1861). Bildung einer zuckerartigen Substanz durch Synthese. *Justus Liebigs Annalen der Chemie* 120(3): 295–298. https://doi.org/10.1002/jlac.18611200308 Aleksandr Butlerov, publishing as Butlerow.

Byrd, A. L., Belkaid, Y., & Segre, J. A. (2018). The human skin microbiome. *Nature Reviews Microbiology* 16(3): 143–155. https://doi.org/10.1038/nrmicro.2017.157

Candi, E., Schmidt, R., & Melino, G. (2005). The cornified envelope: a model of cell death in the skin. *Nature Reviews Molecular Cell Biology* 6(4): 328–340. https://doi.org/10.1038/nrm1619

Chen, I. A., Roberts, R. W., & Szostak, J. W. (2004). The Emergence of Competition Between Model Protocells. *Science* 305(5689): 1474–1476. https://doi.org/10.1126/science.1100757

Christian, H. J., Blakeslee, R. J., Boccippio, D. J., et al. (2003). Global frequency and distribution of lightning as observed from space by the Optical Transient Detector. *Journal of Geophysical Research: Atmospheres* 108(D1): 4005. https://doi.org/10.1029/2002jd002347

Christiansen, M. H., & Kirby, S. (2003). Language evolution: consensus and controversies. *Trends in Cognitive Sciences* 7(7): 300–307. https://doi.org/10.1016/s1364-6613(03)00136-0

Conklin, H. C. (1955). Hanunóo Color Categories. *Southwestern Journal of Anthropology* 11(4): 339–344. https://doi.org/10.1086/soutjanth.11.4.3628909

Cornelis, G., Vernochet, C., Carradec, Q., et al. (2015). Retroviral envelope gene captures and *syncytin* exaptation for placentation in marsupials. *Proceedings of the National Academy of Sciences* 112(5): E487–E496. https://doi.org/10.1073/pnas.1417000112

Crick, F. H. C. (1966). Codon—anticodon pairing: The wobble hypothesis. *Journal of Molecular Biology* 19(2): 548–555. https://doi.org/10.1016/s0022-2836(66)80022-0

Crick, F. H. C. (1968). The origin of the genetic code. *Journal of Molecular Biology* 38(3): 367–379. https://doi.org/10.1016/0022-2836(68)90392-6

Cromartie, H. T., Fonseca, E., Ransom, S. M., et al. (2020). Relativistic Shapiro delay measurements of an extremely massive millisecond pulsar. *Nature Astronomy* 4(1): 72–76. https://doi.org/10.1038/s41550-019-0880-2

Curcio, C. A., & Allen, K. A. (1990). Topography of ganglion cells in human retina. *Journal of Comparative Neurology* 300(1): 5–25. https://doi.org/10.1002/cne.903000103

Curcio, C. A., Sloan, K. R., Kalina, R. E., & Hendrickson, A. E. (1990). Human photoreceptor topography. *Journal of Comparative Neurology* 292(4): 497–523. https://doi.org/10.1002/cne.902920402

Dalrymple, G. B. (2001). The age of the Earth in the twentieth century: a problem (mostly) solved. *Geological Society, London, Special Publications* 190(1): 205–221. https://doi.org/10.1144/gsl.sp.2001.190.01.14

Damer, B., & Deamer, D. (2020). The Hot Spring Hypothesis for an Origin of Life. *Astrobiology* 20(4): 429–452. https://doi.org/10.1089/ast.2019.2045

Danek, A. H., & Wiley, J. (2017). What about False Insights? Deconstructing the Aha! Experience along Its Multiple Dimensions for Correct and Incorrect Solutions Separately. *Frontiers in Psychology* 7: 2077. https://doi.org/10.3389/fpsyg.2016.02077

Darwin, C. (1859). *On the origin of species by means of natural selection, or the preservation of favoured races in the struggle for life.* John Murray.

Dawkins, R. (1976). *The selfish gene.* Oxford University Press.

Deamer, D. W. (1985). Boundary structures are formed by organic components of the Murchison carbonaceous chondrite. *Nature* 317(6040): 792–794. https://doi.org/10.1038/317792a0

Delidovich, I. V., Simonov, A. N., Taran, O. P., & Parmon, V. N. (2014). Catalytic Formation of Monosaccharides: From the Formose Reaction towards Selective Synthesis. *ChemSusChem* 7(7): 1833–1846. https://doi.org/10.1002/cssc.201400040

Demorest, P. B., Pennucci, T., Ransom, S. M., Roberts, M. S. E., & Hessels, J. W. T. (2010). A two-solar-mass neutron star measured using Shapiro delay. *Nature* 467(7319): 1081–1083. https://doi.org/10.1038/nature09466

Dilly, É., Derr, J., & Zanchi, D. (2026). Internal stress controls tendril writhing dynamics in climbing plants. *Proceedings of the National Academy of Sciences* 123(29): e2537010123. https://doi.org/10.1073/pnas.2537010123

Discovery Programme. (2014). *Newgrange Entrance Stone K1* and *Kerbstone 52, Newgrange* [3D models with descriptions]. Sketchfab. https://sketchfab.com/discoveryprogramme

Djokic, T., Van Kranendonk, M. J., Campbell, K. A., Walter, M. R., & Ward, C. R. (2017). Earliest signs of life on land preserved in ca. 3.5 Ga hot spring deposits. *Nature Communications* 8(1): 15263. https://doi.org/10.1038/ncomms15263

Douady, S., & Couder, Y. (1992). Phyllotaxis as a physical self-organized growth process. *Physical Review Letters* 68(13): 2098–2101. https://doi.org/10.1103/physrevlett.68.2098

Dronfield, J. (1995). Subjective vision and the source of Irish megalithic art. *Antiquity* 69(264): 539–549. https://doi.org/10.1017/s0003598x00081928

Ducasse, S., & Langlais, M. (2019). Twenty years on, a new date with Lascaux. Reassessing the chronology of the cave's Paleolithic occupations through new 14C AMS dating. *Paléo* 30(1): 130–147. https://doi.org/10.4000/paleo.4558

Dupressoir, A., Vernochet, C., Bawa, O., et al. (2009). Syncytin-A knockout mice demonstrate the critical role in placentation of a fusogenic, endogenous retrovirus-derived, envelope gene. *Proceedings of the National Academy of Sciences* 106(29): 12127–12132. https://doi.org/10.1073/pnas.0902925106

Eigen, M. (1971). Selforganization of matter and the evolution of biological macromolecules. *Die Naturwissenschaften* 58(10): 465–523. https://doi.org/10.1007/bf00623322

Eigen, M., & Schuster, P. (1977). The hypercycle: A principle of natural self-organization. Part A: Emergence of the hypercycle. *Naturwissenschaften* 64(11): 541–565. https://doi.org/10.1007/bf00450633

Elvehjem, C. A., Madden, R. J., Strong, F. M., & Woolley, D. W. (1937). Relation of Nicotinic Acid and Nicotinic Acid Amide to Canine Black Tongue. *Journal of the American Chemical Society* 59(9): 1767–1768. https://doi.org/10.1021/ja01288a509

Eme, L., Trilles, A., Moreira, D., & Brochier-Armanet, C. (2011). The phylogenomic analysis of the anaphase promoting complex and its targets points to complex and modern-like control of the cell cycle in the last common ancestor of eukaryotes. *BMC Evolutionary Biology* 11(1): 265. https://doi.org/10.1186/1471-2148-11-265

EPICA community members. (2004). Eight glacial cycles from an Antarctic ice core. *Nature* 429(6992): 623–628. https://doi.org/10.1038/nature02599

Esteban, D. J., et al. (2015). Temporal and Spatial Distribution of the Microbial Community of Winogradsky Columns. *PLOS ONE* 10(8): e0134588. https://doi.org/10.1371/journal.pone.0134588

Everett, D. L. (2005). Cultural Constraints on Grammar and Cognition in Pirahã. *Current Anthropology* 46(4): 621–646. https://doi.org/10.1086/431525

Everett, D. L. (2008). *Don't sleep, there are snakes: Life and language in the Amazonian jungle*. Pantheon Books.

Fonseca, E., Cromartie, H. T., Pennucci, T. T., et al. (2021). Refined Mass and Geometric Measurements of the High-mass PSR J0740+6620. *The Astrophysical Journal Letters* 915(1): L12. https://doi.org/10.3847/2041-8213/ac03b8

Food and Agriculture Organization of the United Nations. (1992). *Maize in human nutrition*. FAO. Section 5.2. †

Frank, M. C., Everett, D. L., Fedorenko, E., & Gibson, E. (2008). Number as a cognitive technology: Evidence from Pirahã language and cognition. *Cognition* 108(3): 819–824. https://doi.org/10.1016/j.cognition.2008.04.007

Freeland, S. J., & Hurst, L. D. (1998). The Genetic Code Is One in a Million. *Journal of Molecular Evolution* 47(3): 238–248. https://doi.org/10.1007/pl00006381

Fuller, F. B. (1971). The Writhing Number of a Space Curve. *Proceedings of the National Academy of Sciences* 68(4): 815–819. https://doi.org/10.1073/pnas.68.4.815

Gad-el-Hak, M., Bourgault, D., Riley, J. J., & Chavanne, C. (2025). Comment on "Hidden turbulence in van Gogh's *The Starry Night*". *Physics of Fluids* 37(12): 129101. https://doi.org/10.1063/5.0306945 †

Gerbode, S. J., Puzey, J. R., McCormick, A. G., & Mahadevan, L. (2012). How the Cucumber Tendril Coils and Overwinds. *Science* 337(6098): 1087–1091. https://doi.org/10.1126/science.1223304

Gianni, E., Kwok, S. L. Y., Wan, C. J. K., et al. (2026). A small polymerase ribozyme that can synthesize itself and its complementary strand. *Science* 391(6789): 1022–1028. https://doi.org/10.1126/science.adt2760

Gibson, E., Futrell, R., Jara-Ettinger, J., et al. (2017). Color naming across languages reflects color use. *Proceedings of the National Academy of Sciences* 114(40): 10785–10790. https://doi.org/10.1073/pnas.1619666114

Glory, A. (2008). Lascaux. In A. Glory, B. Delluc, & G. Delluc (Eds.), *Les recherches à Lascaux (1952–1963)* (pp. 23–97). CNRS Éditions. https://doi.org/10.4000/books.editionscnrs.47390 Gallia Préhistoire, supplément 39; published from Glory's manuscript.

Goldberger, J., Waring, C. H., & Willets, D. G. (1915). The Prevention of Pellagra: A Test of Diet among Institutional Inmates. *Public Health Reports* 30(43): 3117–3131. https://doi.org/10.2307/4572932

Gould, S. J. (1996). *Full house: The spread of excellence from Plato to Darwin*. Harmony Books.

Graham, A., & Richardson, J. (2012). Developmental and evolutionary origins of the pharyngeal apparatus. *EvoDevo* 3(1): 24. https://doi.org/10.1186/2041-9139-3-24

Green, J. S. (2010). Feasting with Foam: Ceremonial Drinks of Cacao, Maize, and Pataxte Cacao. In J. E. Staller & M. Carrasco (Eds.), *Pre-Columbian foodways: Interdisciplinary approaches to food, culture, and markets in ancient Mesoamerica* (pp. 315–343). Springer. https://doi.org/10.1007/978-1-4419-0471-3_13

Grice, E. A., & Segre, J. A. (2011). The skin microbiome. *Nature Reviews Microbiology* 9(4): 244–253. https://doi.org/10.1038/nrmicro2537

Guo, N., Hawkins, C., & Nathans, J. (2004). Frizzled6 controls hair patterning in mice. *Proceedings of the National Academy of Sciences* 101(25): 9277–9281. https://doi.org/10.1073/pnas.0402802101

Halder, G., Callaerts, P., & Gehring, W. J. (1995). Induction of Ectopic Eyes by Targeted Expression of the *eyeless* Gene in *Drosophila*. *Science* 267(5205): 1788–1792. https://doi.org/10.1126/science.7892602

Hanczyc, M. M., Fujikawa, S. M., & Szostak, J. W. (2003). Experimental Models of Primitive Cellular Compartments: Encapsulation, Growth, and Division. *Science* 302(5645): 618–622. https://doi.org/10.1126/science.1089904

Hardy, J. L., Frederick, C. M., Kay, P., & Werner, J. S. (2005). Color Naming, Lens Aging, and Grue. *Psychological Science* 16(4): 321–327. https://doi.org/10.1111/j.0956-7976.2005.01534.x

Harper, P. S. (2006). The discovery of the human chromosome number in Lund, 1955–1956. *Human Genetics* 119(1–2): 226–232. https://doi.org/10.1007/s00439-005-0121-x

Hauser, M. D., Yang, C., Berwick, R. C., et al. (2014). The mystery of language evolution. *Frontiers in Psychology* 5: 401. https://doi.org/10.3389/fpsyg.2014.00401

Hofstadter, D. R. (1979). *Gödel, Escher, Bach: An eternal golden braid.* Basic Books.

Hofstadter, D. R. (2007). *I am a strange loop.* Basic Books.

Hordijk, W., & Steel, M. (2004). Detecting autocatalytic, self-sustaining sets in chemical reaction systems. *Journal of Theoretical Biology* 227(4): 451–461. https://doi.org/10.1016/j.jtbi.2003.11.020

Hyde, K. L., Lerch, J., Norton, A., et al. (2009). Musical Training Shapes Structural Brain Development. *The Journal of Neuroscience* 29(10): 3019–3025. https://doi.org/10.1523/jneurosci.5118-08.2009

ICOMOS. (2006). *Rock art of Latin America and the Caribbean* (thematic study). International Council on Monuments and Sites.

Israelachvili, J. N., Mitchell, D. J., & Ninham, B. W. (1976). Theory of self-assembly of hydrocarbon amphiphiles into micelles and bilayers. *Journal of the Chemical Society, Faraday Transactions 2* 72: 1525. https://doi.org/10.1039/f29767201525

Jablonka, E., & Lamb, M. J. (2006). The evolution of information in the major transitions. *Journal of Theoretical Biology* 239(2): 236–246. https://doi.org/10.1016/j.jtbi.2005.08.038 †

Jackson, J. A. (Ed.). (1997). *Glossary of geology* (4th ed.). American Geological Institute.

Johnson, A. P., Cleaves, H. J., Dworkin, J. P., et al. (2008). The Miller Volcanic Spark Discharge Experiment. *Science* 322(5900): 404. https://doi.org/10.1126/science.1161527

Johnson, J. S., Spakowicz, D. J., Hong, B.-Y., et al. (2019). Evaluation of 16S rRNA gene sequencing for species and strain-level microbiome analysis. *Nature Communications* 10(1): 5029. https://doi.org/10.1038/s41467-019-13036-1

Josserand, M., Meeussen, E., Majid, A., & Dediu, D. (2021). Environment and culture shape both the colour lexicon and the genetics of colour perception. *Scientific Reports* 11(1): 19095. https://doi.org/10.1038/s41598-021-98550-3

Jow, D. L., & Scott, D. (2020). Re-evaluating evidence for Hawking points in the CMB. *Journal of Cosmology and Astroparticle Physics* 2020(03): 021. https://doi.org/10.1088/1475-7516/2020/03/021

Kalish, M. L., Griffiths, T. L., & Lewandowsky, S. (2007). Iterated learning: Intergenerational knowledge transmission reveals inductive biases. *Psychonomic Bulletin & Review* 14(2): 288–294. https://doi.org/10.3758/bf03194066

Katz, S. H., Hediger, M. L., & Valleroy, L. A. (1974). Traditional Maize Processing Techniques in the New World. *Science* 184(4138): 765–773. https://doi.org/10.1126/science.184.4138.765 †

Kauffman, S. A. (1986). Autocatalytic sets of proteins. *Journal of Theoretical Biology* 119(1): 1–24. https://doi.org/10.1016/s0022-5193(86)80047-9

Kauffman, S. A. (1993). *The origins of order*. Oxford University Press.

von Kiedrowski, G. (1986). A Self-Replicating Hexadeoxynucleotide. *Angewandte Chemie International Edition in English* 25(10): 932–935. https://doi.org/10.1002/anie.198609322

Kleidon, A. (2012). How does the Earth system generate and maintain thermodynamic disequilibrium and what does it imply for the future of the planet? *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 370(1962): 1012–1040. https://doi.org/10.1098/rsta.2011.0316

Kolmogorov, A. N. (1941/1991). The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences* 434(1890): 9–13. https://doi.org/10.1098/rspa.1991.0075 English translation of the 1941 paper.

Krupnik, I., Aporta, C., Gearheard, S., Laidler, G. J., & Kielsen Holm, L. (Eds.). (2010). *SIKU: Knowing our ice: Documenting Inuit sea ice knowledge and use*. Springer. https://doi.org/10.1007/978-90-481-8587-0

Krupnik, I., & Müller-Wille, L. (2010). Franz Boas and Inuktitut Terminology for Ice and Snow: From the Emergence of the Field to the "Great Eskimo Vocabulary Hoax". In I. Krupnik, C. Aporta, S. Gearheard, G. J. Laidler, & L. Kielsen Holm (Eds.), *SIKU: Knowing our ice* (pp. 377–400). Springer. https://doi.org/10.1007/978-90-481-8587-0_16

Lai, Y.-C., Liu, Z., & Chen, I. A. (2021). Encapsulation of ribozymes inside model protocells leads to faster evolutionary adaptation. *Proceedings of the National Academy of Sciences* 118(21): e2025054118. https://doi.org/10.1073/pnas.2025054118

Lane, N., Allen, J. F., & Martin, W. (2010). How did LUCA make a living? Chemiosmosis in the origin of life. *BioEssays* 32(4): 271–280. https://doi.org/10.1002/bies.200900131

Lane, N. (2015). *The vital question: Why is life the way it is?* Profile Books.

Laukkonen, R. E., Kaveladze, B. T., Tangen, J. M., & Schooler, J. W. (2020). The dark side of Eureka: Artificially induced Aha moments make facts feel true. *Cognition* 196: 104122. https://doi.org/10.1016/j.cognition.2019.104122

Lavialle, C., Cornelis, G., Dupressoir, A., et al. (2013). Paleovirology of '*syncytins*', retroviral *env* genes exapted for a role in placentation. *Philosophical Transactions of the Royal Society B: Biological Sciences* 368(1626): 20120507. https://doi.org/10.1098/rstb.2012.0507

Lee, D. H., Granja, J. R., Martinez, J. A., Severin, K., & Ghadiri, M. R. (1996). A self-replicating peptide. *Nature* 382(6591): 525–528. https://doi.org/10.1038/382525a0

Levinson, S. C. (2003). *Space in language and cognition*. Cambridge University Press.

Lewis-Williams, J. D., & Dowson, T. A. (1988). The Signs of All Times: Entoptic Phenomena in Upper Palaeolithic Art. *Current Anthropology* 29(2): 201–245. https://doi.org/10.1086/203629

Lin, C. C., & Shu, F. H. (1964). On the Spiral Structure of Disk Galaxies. *The Astrophysical Journal* 140: 646–655. https://doi.org/10.1086/147955

Lincoln, T. A., & Joyce, G. F. (2009). Self-Sustained Replication of an RNA Enzyme. *Science* 323(5918): 1229–1232. https://doi.org/10.1126/science.1167856

Lindsey, D. T., & Brown, A. M. (2002). Color Naming and the Phototoxic Effects of Sunlight on the Eye. *Psychological Science* 13(6): 506–512. https://doi.org/10.1111/1467-9280.00489

Liu, L. F., & Wang, J. C. (1987). Supercoiling of the DNA template during transcription. *Proceedings of the National Academy of Sciences* 84(20): 7024–7027. https://doi.org/10.1073/pnas.84.20.7024

Luger, K., Mäder, A. W., Richmond, R. K., Sargent, D. F., & Richmond, T. J. (1997). Crystal structure of the nucleosome core particle at 2.8 Å resolution. *Nature* 389(6648): 251–260. https://doi.org/10.1038/38444

Luo, Z.-X. (2011). Developmental Patterns in Mesozoic Evolution of Mammal Ears. *Annual Review of Ecology, Evolution, and Systematics* 42(1): 355–380. https://doi.org/10.1146/annurev-ecolsys-032511-142302 †

Lüthi, D., Le Floch, M., Bereiter, B., et al. (2008). High-resolution carbon dioxide concentration record 650,000–800,000 years before present. *Nature* 453(7193): 379–382. https://doi.org/10.1038/nature06949

Ma, Y., Cheng, W., Huang, S., et al. (2024). Hidden turbulence in van Gogh's *The Starry Night*. *Physics of Fluids* 36(9): 095140. https://doi.org/10.1063/5.0213627

Madison, K. C. (2003). Barrier Function of the Skin: "La Raison d'Être" of the Epidermis. *Journal of Investigative Dermatology* 121(2): 231–241. https://doi.org/10.1046/j.1523-1747.2003.12359.x †

Marcot, J. D., & McShea, D. W. (2007). Increasing hierarchical complexity throughout the history of life: phylogenetic tests of trend

mechanisms. *Paleobiology* 33(2): 182–200. https://doi.org/10.1666/06028.1

Margulis, L. (1995). Gaia is a tough bitch. In J. Brockman, *The third culture: Beyond the scientific revolution* (ch. 7). Simon & Schuster.

Martin, L. (1986). "Eskimo Words for Snow": A Case Study in the Genesis and Decay of an Anthropological Example. *American Anthropologist* 88(2): 418–423. https://doi.org/10.1525/aa.1986.88.2.02a00080

Martin, W., & Russell, M. J. (2003). On the origins of cells: a hypothesis for the evolutionary transitions from abiotic geochemistry to chemoautotrophic prokaryotes, and from prokaryotes to nucleated cells. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* 358(1429): 59–85. https://doi.org/10.1098/rstb.2002.1183

Martinovic, J., Paramei, G. V., & MacInnes, W. J. (2020). Russian blues reveal the limits of language influencing colour discrimination. *Cognition* 201: 104281. https://doi.org/10.1016/j.cognition.2020.104281

Massey, S. E. (2008). A Neutral Origin for Error Minimization in the Genetic Code. *Journal of Molecular Evolution* 67(5): 510–516. https://doi.org/10.1007/s00239-008-9167-4

Matsumura, S., Kun, Á., Ryckelynck, M., et al. (2016). Transient compartmentalization of RNA replicators prevents extinction due to parasites. *Science* 354(6317): 1293–1296. https://doi.org/10.1126/science.aag1582

Maynard Smith, J., & Szathmáry, E. (1995). *The major transitions in evolution*. W. H. Freeman Spektrum.

Maynard Smith, J., & Szathmáry, E. (1999). *The origins of life*. Oxford University Press.

McFall-Ngai, M., Hadfield, M. G., Bosch, T. C. G., et al. (2013). Animals in a bacterial world, a new imperative for the life sciences. *Proceedings of the National Academy of Sciences* 110(9): 3229–3236. https://doi.org/10.1073/pnas.1218525110

McShea, D. W. (1996). Metazoan Complexity and Evolution: Is There a Trend? *Evolution* 50(2): 477–492. https://doi.org/10.1111/j.1558-5646.1996.tb03861.x

McShea, D. W., & Simpson, C. (2011). The Miscellaneous Transitions in Evolution. In B. Calcott & K. Sterelny (Eds.), *The major transitions in evolution revisited* (pp. 19–34). MIT Press. https://doi.org/10.7551/mitpress/9780262015240.003.0002

Mesoudi, A., Whiten, A., & Dunbar, R. (2006). A bias for social information in human cultural transmission. *British Journal of Psychology* 97(3): 405–423. https://doi.org/10.1348/000712605x85871

Mesoudi, A., & Whiten, A. (2008). The multiple roles of cultural transmission experiments in understanding human cultural evolution. *Philosophical Transactions of the Royal Society B: Biological Sciences* 363(1509): 3489–3501. https://doi.org/10.1098/rstb.2008.0129

Mi, S., Lee, X., Li, X.-P., et al. (2000). Syncytin is a captive retroviral envelope protein involved in human placental morphogenesis. *Nature* 403(6771): 785–789. https://doi.org/10.1038/35001608

Miller, S. L. (1953). A Production of Amino Acids Under Possible Primitive Earth Conditions. *Science* 117(3046): 528–529. https://doi.org/10.1126/science.117.3046.528

Minsky, M. (1986). *The society of mind*. Simon & Schuster.

Moody, E. R. R., Álvarez-Carretero, S., Mahendrarajah, T. A., et al. (2024). The nature of the last universal common ancestor and its impact on the early Earth system. *Nature Ecology & Evolution* 8(9): 1654–1666. https://doi.org/10.1038/s41559-024-02461-1

Naik, S., Bouladoux, N., Wilhelm, C., et al. (2012). Compartmentalized Control of Skin Immunity by Resident Commensals. *Science* 337(6098): 1115–1119. https://doi.org/10.1126/science.1225152

Nasim, O. W. (2010). Observation, working images and procedure: the 'Great Spiral' in Lord Rosse's astronomical record books and

beyond. *The British Journal for the History of Science* 43(3): 353–389. https://doi.org/10.1017/s0007087410000051

Navarro-González, R., McKay, C. P., & Mvondo, D. N. (2001). A possible nitrogen crisis for Archaean life due to reduced nitrogen fixation by lightning. *Nature* 412(6842): 61–64. https://doi.org/10.1038/35083537

Nevins, A., Pesetsky, D., & Rodrigues, C. (2009). Pirahã exceptionality: A reassessment. *Language* 85(2): 355–404. https://doi.org/10.1353/lan.0.0107

Newman, E. B., Mark, R. K., & Vivian, R. G. (1982). Anasazi Solar Marker: The Use of a Natural Rockfall. *Science* 217(4564): 1036–1038. https://doi.org/10.1126/science.217.4564.1036

Norenzayan, A., Atran, S., Faulkner, J., & Schaller, M. (2006). Memory and Mystery: The Cultural Selection of Minimally Counterintuitive Narratives. *Cognitive Science* 30(3): 531–553. https://doi.org/10.1207/s15516709cog0000_68

O'Kelly, M. J. (1982). *Newgrange: Archaeology, art and legend*. Thames and Hudson. †

O'Malley, M. A., & Powell, R. (2016). Major problems in evolutionary transitions: how a metabolic perspective can enrich our understanding of macroevolution. *Biology & Philosophy* 31(2): 159–189. https://doi.org/10.1007/s10539-015-9513-z

Orgel, L. E. (2008). The Implausibility of Metabolic Cycles on the Prebiotic Earth. *PLoS Biology* 6(1): e18. https://doi.org/10.1371/journal.pbio.0060018

Ortiz, F. (1947). *El huracán: Su mitología y sus símbolos*. Fondo de Cultura Económica.

Penrose, R. (2010). *Cycles of time: An extraordinary new view of the universe*. Bodley Head.

Philippe, N., Legendre, M., Doutre, G., et al. (2013). Pandoraviruses: Amoeba Viruses with Genomes Up to 2.5 Mb Reaching That of Parasitic Eukaryotes. *Science* 341(6143): 281–286. https://doi.org/10.1126/science.1239181

Prigogine, I. (1978). Time, Structure, and Fluctuations. *Science* 201(4358): 777–785. https://doi.org/10.1126/science.201.4358.777 Nobel lecture, delivered 8 December 1977.

Prusiner, S. B. (1982). Novel Proteinaceous Infectious Particles Cause Scrapie. *Science* 216(4542): 136–144. https://doi.org/10.1126/science.6801762

Pullum, G. K. (1989). The Great Eskimo Vocabulary Hoax. *Natural Language and Linguistic Theory* 7(2): 275–281. https://doi.org/10.1007/bf00138079 Published in the journal's "Topic…Comment" column.

Pullum, G. K. (1991). *The great Eskimo vocabulary hoax and other irreverent essays on the study of language*. University of Chicago Press.

Rajakumar, K. (2000). Pellagra in the United States. *Southern Medical Journal* 93(3): 272–277. https://doi.org/10.1097/00007611-200093030-00005 †

Rajamani, S., Vlassov, A., Benner, S., et al. (2008). Lipid-assisted Synthesis of RNA-like Polymers from Mononucleotides. *Origins of Life and Evolution of Biospheres* 38(1): 57–74. https://doi.org/10.1007/s11084-007-9113-2

Ranjan, S., Todd, Z. R., Sutherland, J. D., & Sasselov, D. D. (2018). Sulfidic Anion Concentrations on Early Earth for Surficial Origins-of-Life Chemistry. *Astrobiology* 18(8): 1023–1040. https://doi.org/10.1089/ast.2017.1770

Ranjan, S., Todd, Z. R., Rimmer, P. B., Sasselov, D. D., & Babbin, A. R. (2019). Nitrogen Oxide Concentrations in Natural Waters on Early Earth. *Geochemistry, Geophysics, Geosystems* 20(4): 2021–2039. https://doi.org/10.1029/2018gc008082

Raoult, D., Audic, S., Robert, C., et al. (2004). The 1.2-Megabase Genome Sequence of Mimivirus. *Science* 306(5700): 1344–1350. https://doi.org/10.1126/science.1101485

Ratnieks, F. L. W., & Visscher, P. K. (1989). Worker policing in the honeybee. *Nature* 342(6251): 796–797. https://doi.org/10.1038/342796a0

Ricardo, A., Carrigan, M. A., Olcott, A. N., & Benner, S. A. (2004). Borate Minerals Stabilize Ribose. *Science* 303(5655): 196. https://doi.org/10.1126/science.1092464

Riley, J. J., & Gad-el-Hak, M. (2025). Is there hidden turbulence in Vincent van Gogh's *The Starry Night*? *Journal of Turbulence* 26(4): 113–114. https://doi.org/10.1080/14685248.2025.2477244 †

Robock, A. (2000). Volcanic eruptions and climate. *Reviews of Geophysics* 38(2): 191–219. https://doi.org/10.1029/1998rg000054 †

Robson, D. (2012). Chilly words: how Eskimos really say "snow". *New Scientist* 216(2896): 72–73. https://doi.org/10.1016/s0262-4079(12)63275-8 †

Romani, R. W., Kandel, D., Filippenko, A. V., Brink, T. G., & Zheng, W. (2022). PSR J0952–0607: The Fastest and Heaviest Known Galactic Neutron Star. *The Astrophysical Journal Letters* 934(2): L17. https://doi.org/10.3847/2041-8213/ac8007

Rundell, E. A., Banta, L. M., Ward, D. V., et al. (2014). 16S rRNA Gene Survey of Microbial Communities in Winogradsky Columns. *PLoS ONE* 9(8): e104134. https://doi.org/10.1371/journal.pone.0104134

Sagan, L. (1967). On the origin of mitosing cells. *Journal of Theoretical Biology* 14(3): 225–274. https://doi.org/10.1016/0022-5193(67)90079-3 Lynn Margulis, publishing as Lynn Sagan.

Salvi, C., Bricolo, E., Kounios, J., Bowden, E., & Beeman, M. (2016). Insight solutions are correct more often than analytic solutions. *Thinking & Reasoning* 22(4): 443–460. https://doi.org/10.1080/13546783.2016.1141798

Salvini-Plawen, L. V., & Mayr, E. (1977). On the Evolution of Photoreceptors and Eyes. *Evolutionary Biology* 10: 207–263. https://doi.org/10.1007/978-1-4615-6953-4_4

Samuel, S., Cole, G., & Eacott, M. J. (2019). Grammatical gender and linguistic relativity: A systematic review. *Psychonomic Bulletin & Review* 26(6): 1767–1786. https://doi.org/10.3758/s13423-019-01652-3

Santini, L. M., Weber, S. L., Marston, J. M., & Runggaldier, A. (2022). First archaeological identification of nixtamalized maize, from two pit latrines at the ancient Maya site of San Bartolo, Guatemala. *Journal of Archaeological Science* 143: 105581. https://doi.org/10.1016/j.jas.2022.105581

Schrödinger, E. (1944). *What is life? The physical aspect of the living cell*. Cambridge University Press.

Schumann, W. O. (1952). Über die strahlungslosen Eigenschwingungen einer leitenden Kugel, die von einer Luftschicht und einer Ionosphärenhülle umgeben ist. *Zeitschrift für Naturforschung A* 7(2): 149–154. https://doi.org/10.1515/zna-1952-0202

Schwartz, U., Németh, A., Diermeier, S., et al. (2019). Characterizing the nuclease accessibility of DNA in human cells to map higher order structures of chromatin. *Nucleic Acids Research* 47(3): 1239–1254. https://doi.org/10.1093/nar/gky1203

Sender, R., Fuchs, S., & Milo, R. (2016). Revised Estimates for the Number of Human and Bacteria Cells in the Body. *PLOS Biology* 14(8): e1002533. https://doi.org/10.1371/journal.pbio.1002533

Shubin, N. (2008). *Your inner fish*. Pantheon.

Smith, H. F., Fisher, R. E., Everett, M. L., et al. (2009). Comparative anatomy and phylogenetic distribution of the mammalian cecal appendix. *Journal of Evolutionary Biology* 22(10): 1984–1999. https://doi.org/10.1111/j.1420-9101.2009.01809.x

Smolin, L. (1992). Did the Universe evolve? *Classical and Quantum Gravity* 9(1): 173–191. https://doi.org/10.1088/0264-9381/9/1/016

Smolin, L. (1997). *The life of the cosmos*. Oxford University Press.

Smolin, L. (2007). Scientific alternatives to the anthropic principle. In B. Carr (Ed.), *Universe or multiverse?* (pp. 323–366). Cambridge University Press. https://doi.org/10.1017/cbo9781107050990.022

Smyth, J., Carlin, N., Hofmann, D., et al. (2025). The 'king' of Newgrange? A critical analysis of a Neolithic petrous fragment from the passage tomb chamber. *Antiquity* 99(405): 672–688. https://doi.org/10.15184/aqy.2025.63

Sofaer, A., Zinser, V., & Sinclair, R. M. (1979). A Unique Solar Marking Construct. *Science* 206(4416): 283–291. https://doi.org/10.1126/science.206.4416.283

Szathmáry, E. (2015). Toward major evolutionary transitions theory 2.0. *Proceedings of the National Academy of Sciences* 112(33): 10104–10111. https://doi.org/10.1073/pnas.1421398112

Thompson, D. W. (1917). *On growth and form.* Cambridge University Press.

Timmis, J. N., Ayliffe, M. A., Huang, C. Y., & Martin, W. (2004). Endosymbiotic gene transfer: organelle genomes forge eukaryotic chromosomes. *Nature Reviews Genetics* 5(2): 123–135. https://doi.org/10.1038/nrg1271

Trifonov, E. N. (2011). Vocabulary of Definitions of Life Suggests a Definition. *Journal of Biomolecular Structure and Dynamics* 29(2): 259–266. https://doi.org/10.1080/073911011010524992

Turing, A. M. (1952). The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* 237(641): 37–72. https://doi.org/10.1098/rstb.1952.0012

Vinograd, J., Lebowitz, J., Radloff, R., Watson, R., & Laipis, P. (1965). The twisted circular form of polyoma viral DNA. *Proceedings of the National Academy of Sciences* 53(5): 1104–1111. https://doi.org/10.1073/pnas.53.5.1104

Wächtershäuser, G. (1988). Before enzymes and templates: theory of surface metabolism. *Microbiological Reviews* 52(4): 452–484. https://doi.org/10.1128/mr.52.4.452-484.1988

Wang, J. C. (2002). Cellular roles of DNA topoisomerases: a molecular perspective. *Nature Reviews Molecular Cell Biology* 3(6): 430–440. https://doi.org/10.1038/nrm831

Wedel, M. J. (2012). A Monument of Inefficiency: The Presumed Course of the Recurrent Laryngeal Nerve in Sauropod Dinosaurs. *Acta Palaeontologica Polonica* 57(2): 251–256. https://doi.org/10.4202/app.2011.0019

Weisburg, W. G., Barns, S. M., Pelletier, D. A., & Lane, D. J. (1991). 16S ribosomal DNA amplification for phylogenetic study. *Journal of Bacteriology* 173(2): 697–703. https://doi.org/10.1128/jb.173.2.697-703.1991

Weiss, M. C., Sousa, F. L., Mrnjavac, N., et al. (2016). The physiology and habitat of the last universal common ancestor. *Nature Microbiology* 1(9): 16116. https://doi.org/10.1038/nmicrobiol.2016.116

Whitney, C. A. (1986). The Skies of Vincent van Gogh. *Art History* 9(3): 351–362. https://doi.org/10.1111/j.1467-8365.1986.tb00206.x †

Whitney, C. A. (1992). 'The Starry Night' Illuminated. *Physics Today* 45(8): 13. https://doi.org/10.1063/1.2809764 A letter.

Whorf, B. L. (1940). Science and linguistics. *Technology Review* 42: 229–231.

Williams, G. E. (2000). Geological constraints on the Precambrian history of Earth's rotation and the Moon's orbit. *Reviews of Geophysics* 38(1): 37–59. https://doi.org/10.1029/1999rg900016

Winawer, J., Witthoft, N., Frank, M. C., et al. (2007). Russian blues reveal effects of language on color discrimination. *Proceedings of the National Academy of Sciences* 104(19): 7780–7785. https://doi.org/10.1073/pnas.0701644104

Winfree, A. T. (1972). Spiral Waves of Chemical Activity. *Science* 175(4022): 634–636. https://doi.org/10.1126/science.175.4022.634

Woese, C. R., & Fox, G. E. (1977). Phylogenetic structure of the prokaryotic domain: The primary kingdoms. *Proceedings of the National Academy of Sciences* 74(11): 5088–5090. https://doi.org/10.1073/pnas.74.11.5088

Woese, C. R., Kandler, O., & Wheelis, M. L. (1990). Towards a natural system of organisms: proposal for the domains Archaea,

Bacteria, and Eucarya. *Proceedings of the National Academy of Sciences* 87(12): 4576–4579. https://doi.org/10.1073/pnas.87.12.4576

Woese, C. R. (2002). On the evolution of cells. *Proceedings of the National Academy of Sciences* 99(13): 8742–8747. https://doi.org/10.1073/pnas.132266999

Zhu, T. F., & Szostak, J. W. (2009). Coupled Growth and Division of Model Protocell Membranes. *Journal of the American Chemical Society* 131(15): 5705–5713. https://doi.org/10.1021/ja900919c

About the author

Freddi Van Gogh grew up moving between England, Germany, and the United States, often left to themselves, out in nature or in an empty Oxbridge library, with a great deal of curiosity and a habit of getting lost in books.

Freddi lives a mostly private life, doing science by day and writing fantasy by night. They are not related to "the famous Van Goghs", but have been taken for a lost cousin often enough that they stopped correcting people.

Freddi can be reached at freddi.van.gogh@proton.me, but values their privacy, and so writes under a changed first name.

Natural Knowledge - a collection

1. *Natural Knowledge* — main edition
   How complexity accumulates, from prebiotic chemistry through life, mind, language, and culture, with every claim marked as established, contested, or the author's own.
2. *Natural Knowledge* — high school edition (this book)
   The same account, shorter and plainer, for younger readers, with each idea boxed as settled, still argued about, or the author's own.
3. *Modern Alchemy*
   The book's concepts as a numbered glossary with a notation for combining them, and a frank list of what the notation cannot say. It opens with a primer on alchemy.
4. *Prolog Logicbase*
   The fact checker behind the series: the books' key claims written as logic and tested for contradictions and overstatement.